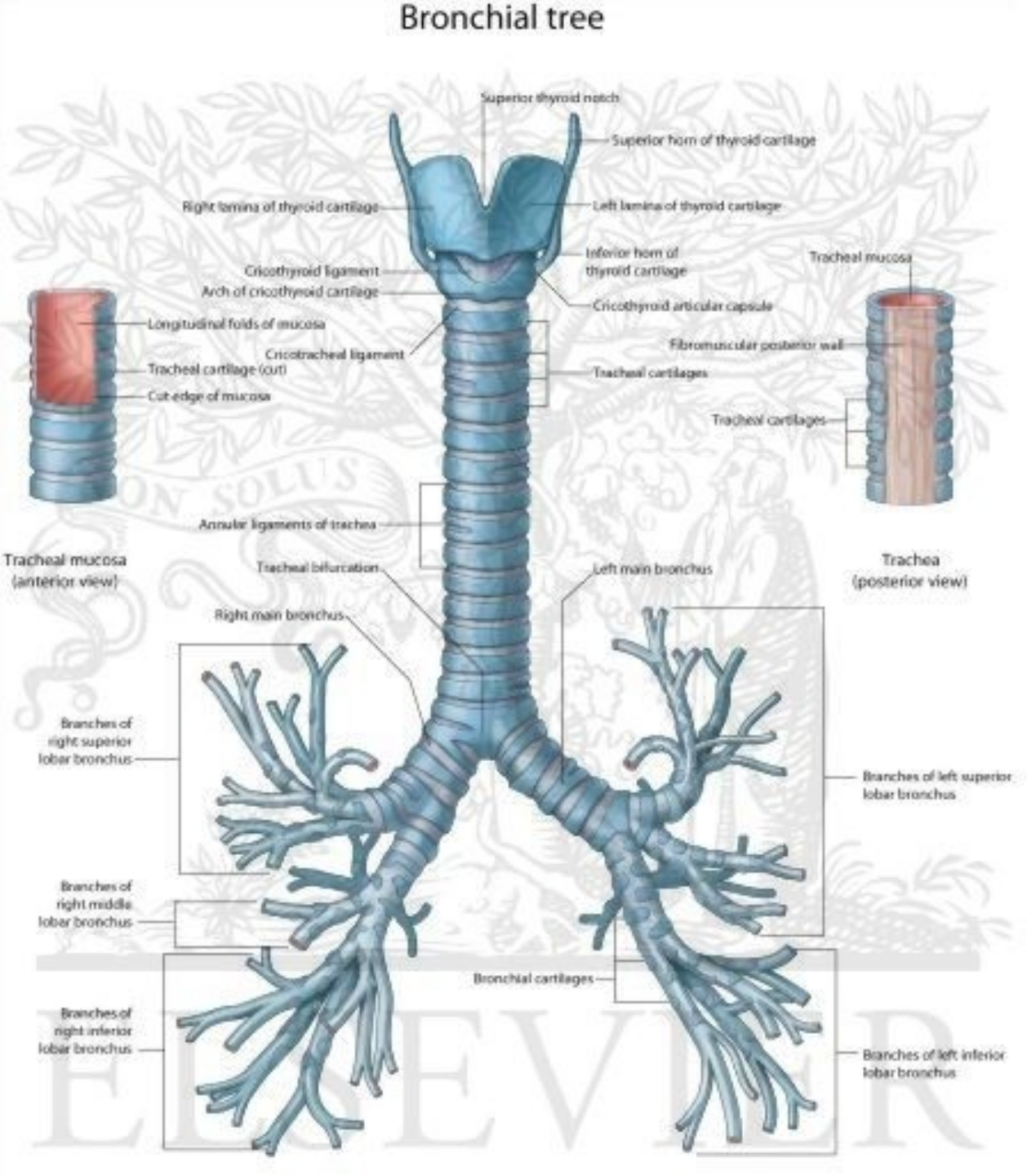
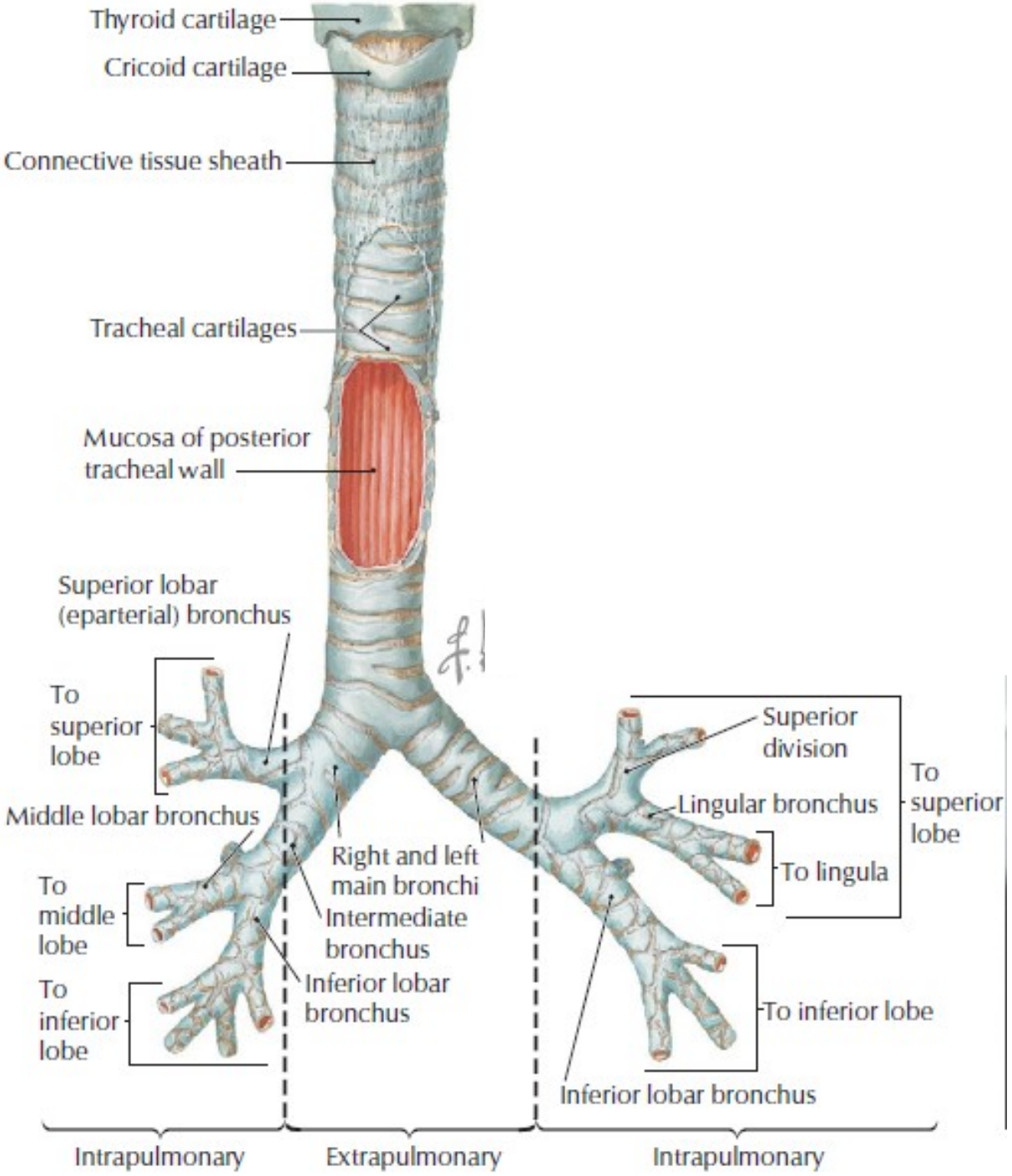
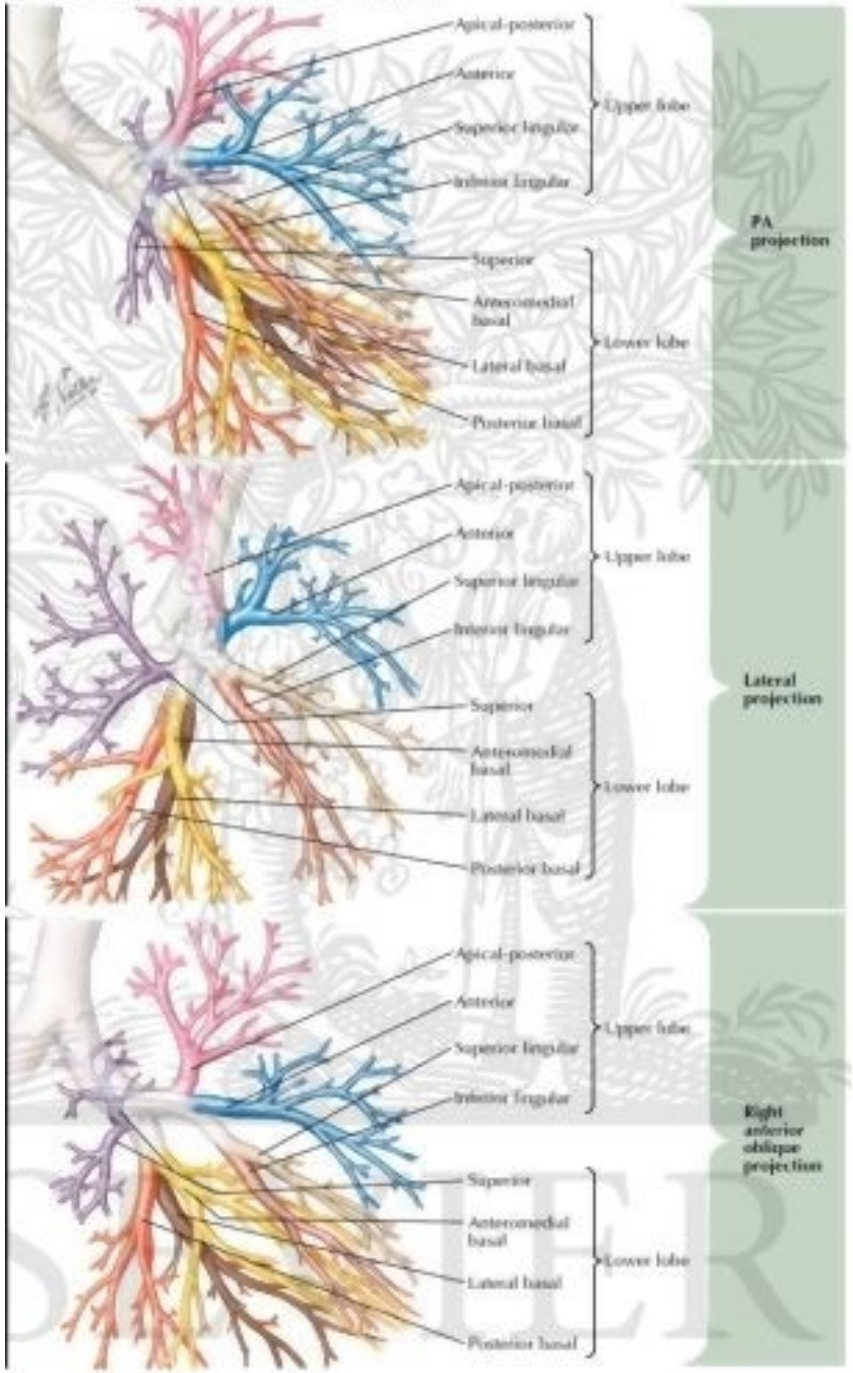

بسم الله الرحمن الرحيم
١٤٣٠

In the lungs we find the last segments of the conductive portion of the respiratory system. The main bronchi divide into lobar bronchi which in turn give rise to segmental bronchi. The latter supply the bronchopulmonary segments of the lungs.




Thyroid cartilage
Cricoid cartilage
Connective tissue sheath
Tracheal cartilages
Mucosa of posterior tracheal wall
Superior lobar (eparterial) bronchus
To superior lobe
Middle lobar bronchus
To middle lobe
To inferior lobe
Right and left main bronchi
Intermediate bronchus
Inferior lobar bronchus
Superior division
Lingular bronchus
To lingula
To superior lobe
To inferior lobe
Inferior lobar bronchus
Intrapulmonary
Extrapulmonary
Intrapulmonary

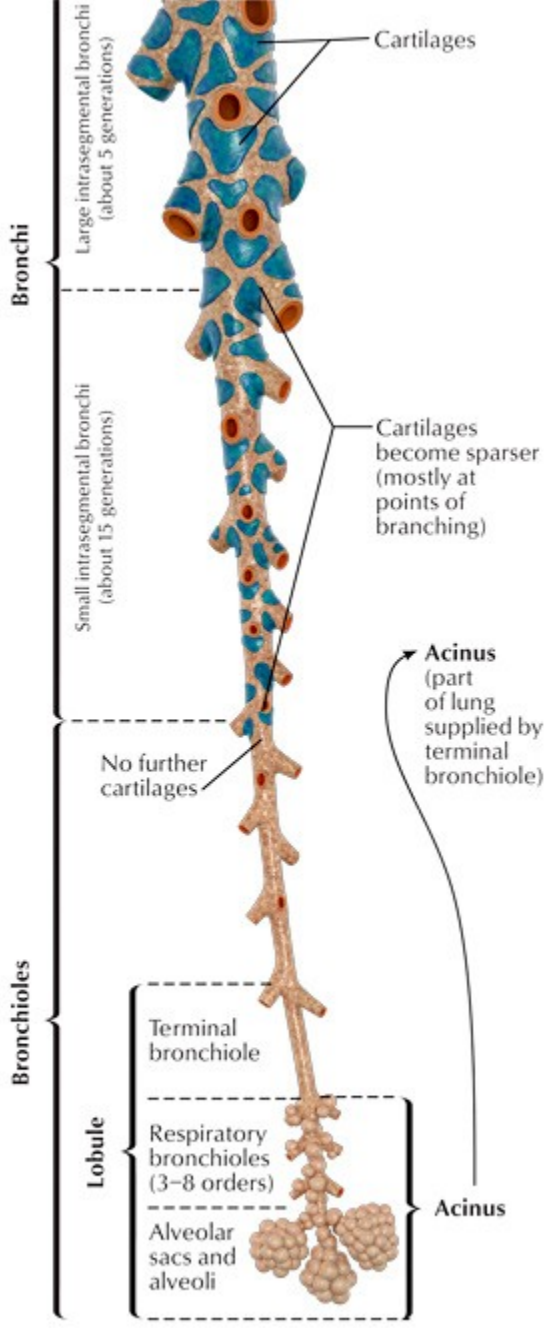
Left Bronchial Tree as Revealed by Bronchograms



After 10 to 20 generations of divisions of *bronchi* the *bronchioles* are formed with 1mm diameter tube.

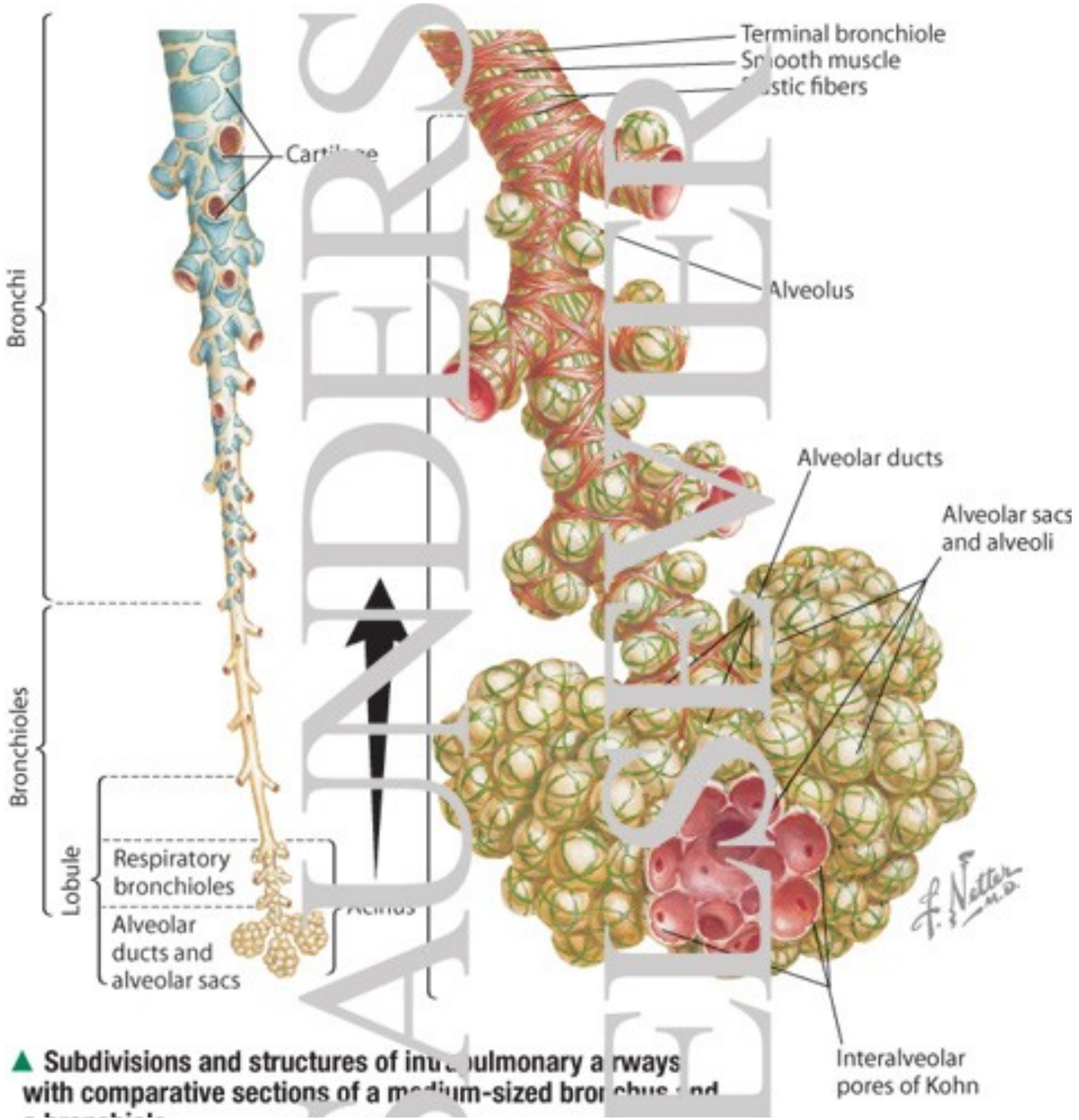
Each *bronchiole* branches further to give rise 5 to 7 *terminal bronchioles*.

Each *terminal bronchiole* branches into two *respiratory bronchioles*.

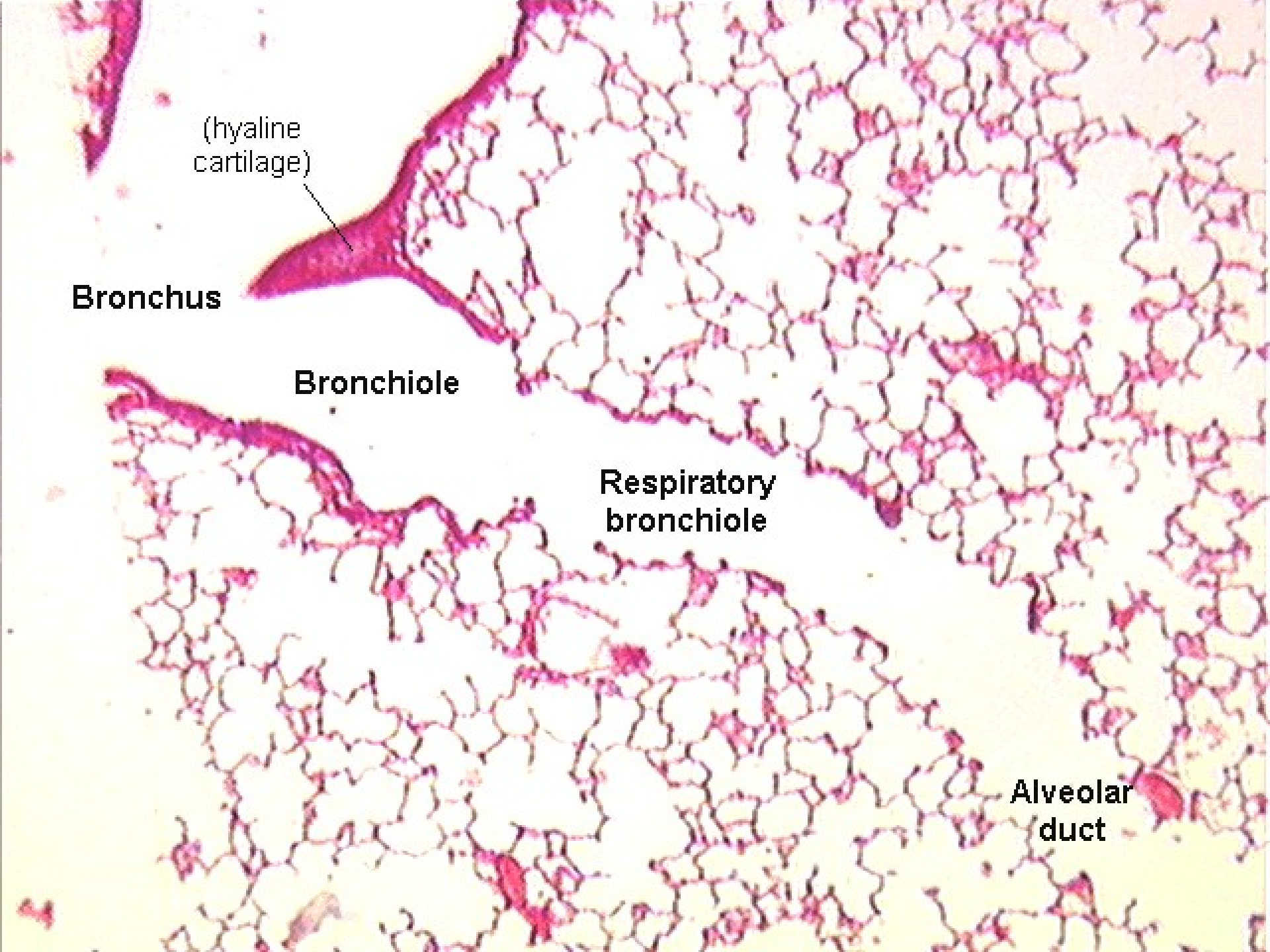


Subdivisions of intrapulmonary airways

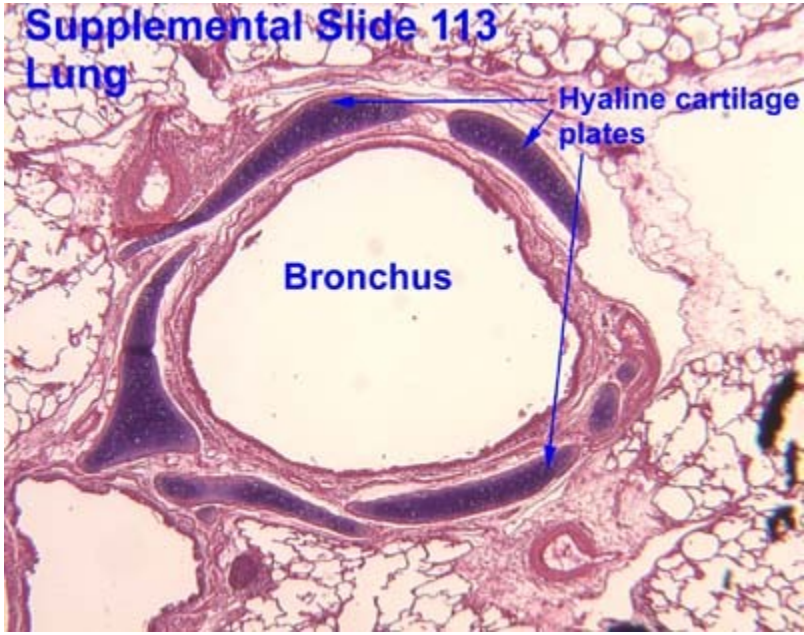
respiratory bronchioles which divides further to give rise *alveolar ducts*. These ducts leads to into *pulmonary atria* which open into *alveolar sacs* & *pulmonary alveoli*.

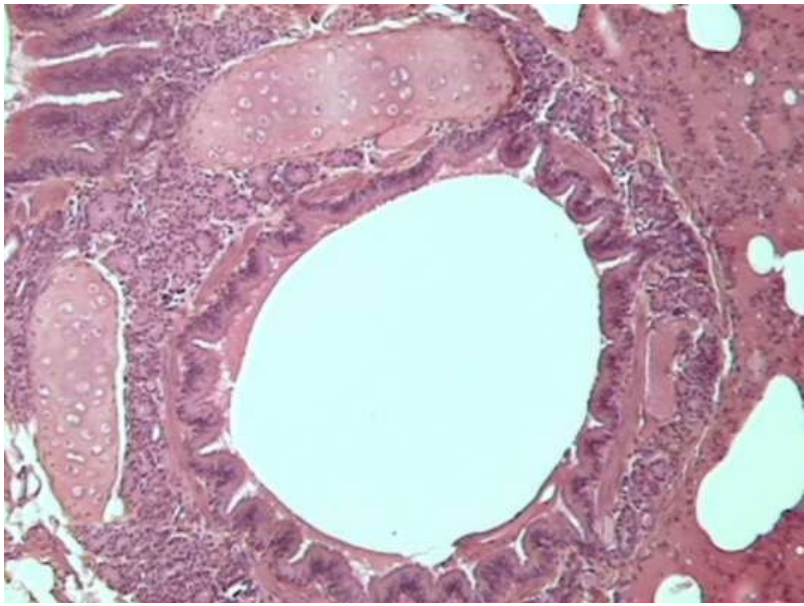


▲ Subdivisions and structures of intrapulmonary airways with comparative sections of a medium-sized bronchus and a bronchiole


(hyaline
cartilage)
Bronchus
Bronchiole
Respiratory
bronchiole
Alveolar
duct

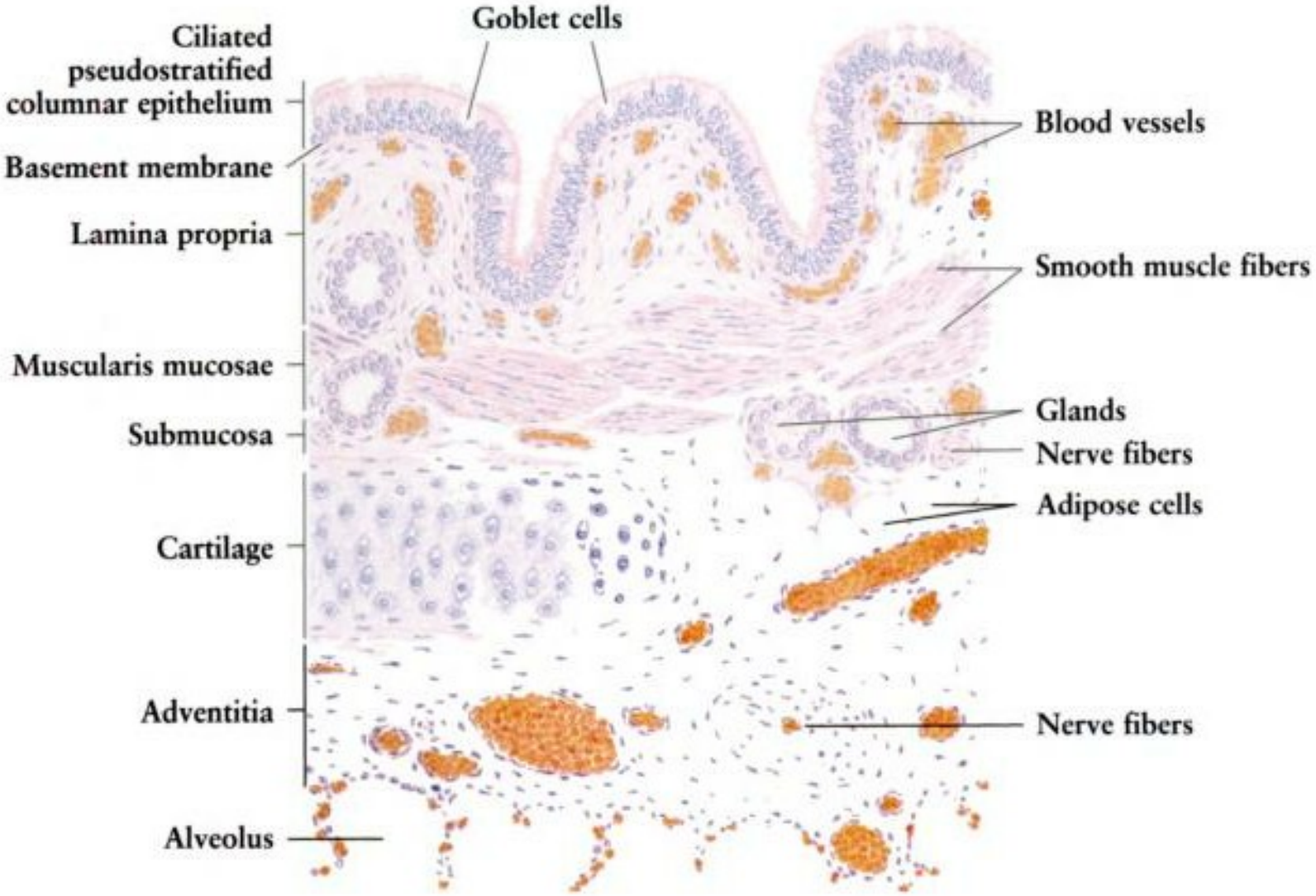
- **Bronchi**
- The extrapulmonary bronchi is structurally similar to the tracheal structure except for their smaller diameter.
- The intrapulmonary bronchi is differ from extrapulmonary bronchi in many respects that intrapulmonary bronchi have smaller diameter & round contour, smooth muscle appear in form of a complete layer & hyaline cartilage is present as irregular cartilaginous plates.
- In addition, bronchi are surrounded by a layer of smooth muscle, which is located between the cartilage and epithelium.
- Bronchi are characterized by the presence of glands and supporting cartilage.
- In the primary bronchi most cartilage rings completely encircle the lumen, but as the bronchial diameter decreases, cartilage rings are gradually replaced with isolated plates of hyaline cartilage (insulae cartilagenea)
- Abundant mucous and serous glands are also present, with ducts opening into the bronchial lumen.
- Mucosa lined by pseudostratified columnar ciliated epithelium with goblet cells.
- The lamina propria contains elastic fibers and abundant mucous and serous glands, whose ducts open into the bronchial lumen.
- Numerous lymphocytes are found both within the lamina propria and among the epithelial cells.
- In the bronchial lamina propria, is a layer of crisscrossing bundles of spirally arranged smooth muscle, which become more prominent in the smaller bronchial branches. Contraction of this muscle layer is responsible for the folded





Lung: Small (Intrapulmonary) Bronchus

Human • H.E. stain • Medium magnification

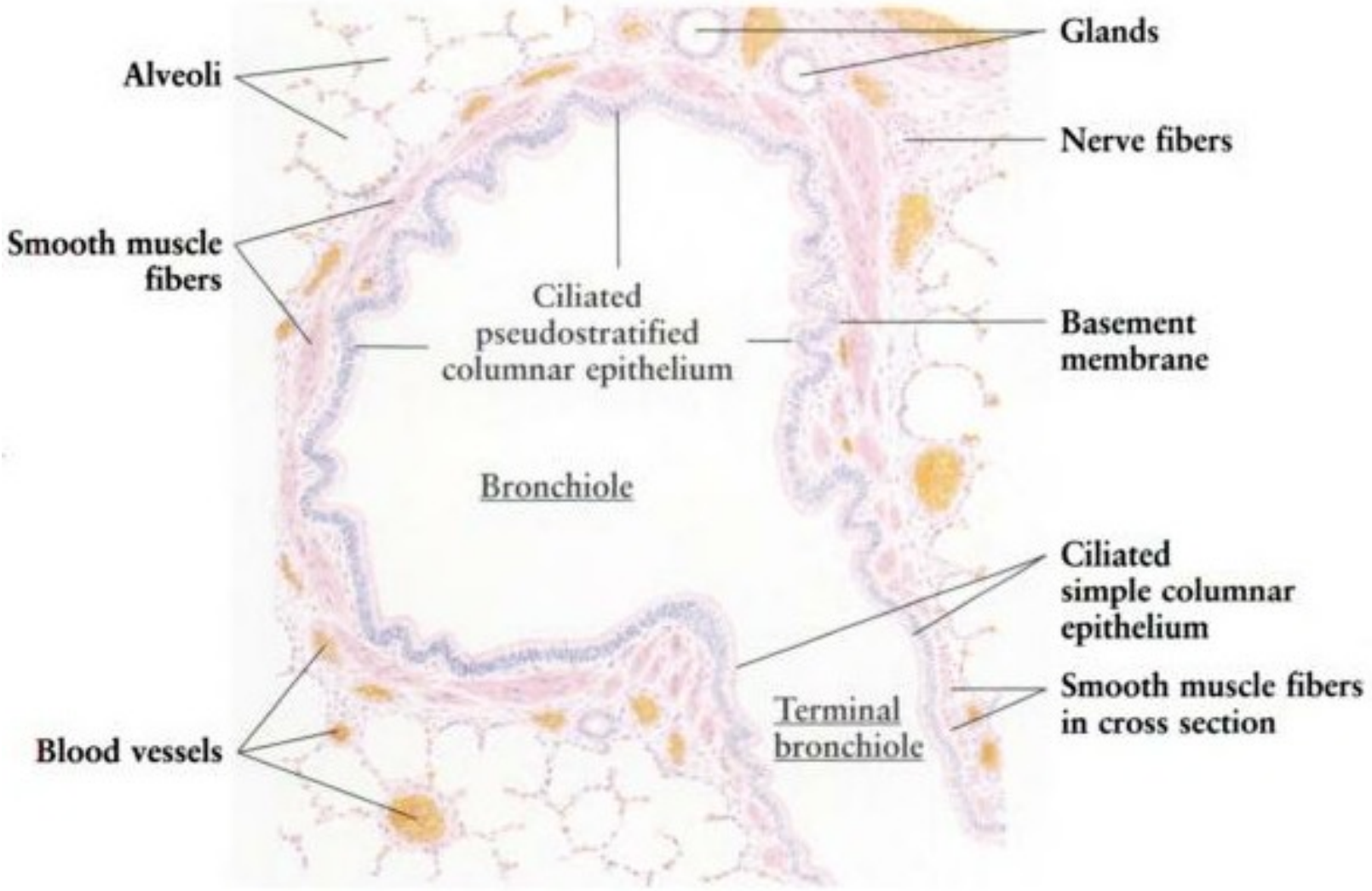


- **Bronchioles**
- Bronchioles are the terminal segments of the conductive portion.
- Bronchioles are the intralobular airways with diameters of 1 mm or less, formed after about the tenth generation of branching, and have neither cartilage nor glands in their mucosa.
- At the transition from bronchi to bronchioles the epithelium changes to a ciliated columnar epithelium, but most of the cell types found in the epithelium of other parts of the conductive portion are still present.
- The layer of smooth muscle is relatively thicker than in the bronchi.
- In the larger bronchioles, the epithelium is still ciliated pseudostratified columnar epithelium showing many goblet cells , but this decreases in height and complexity to become ciliated simple columnar & then ciliated simple cuboidal epithelium in the

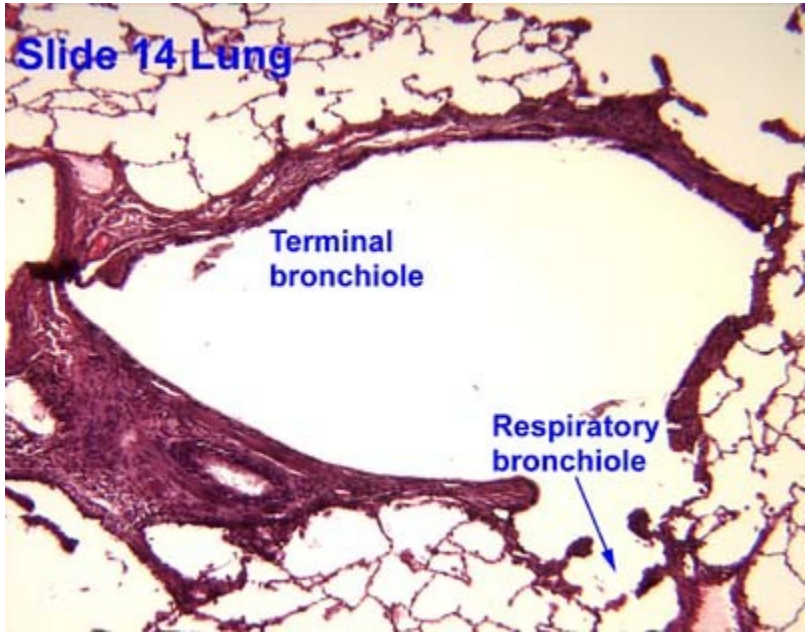


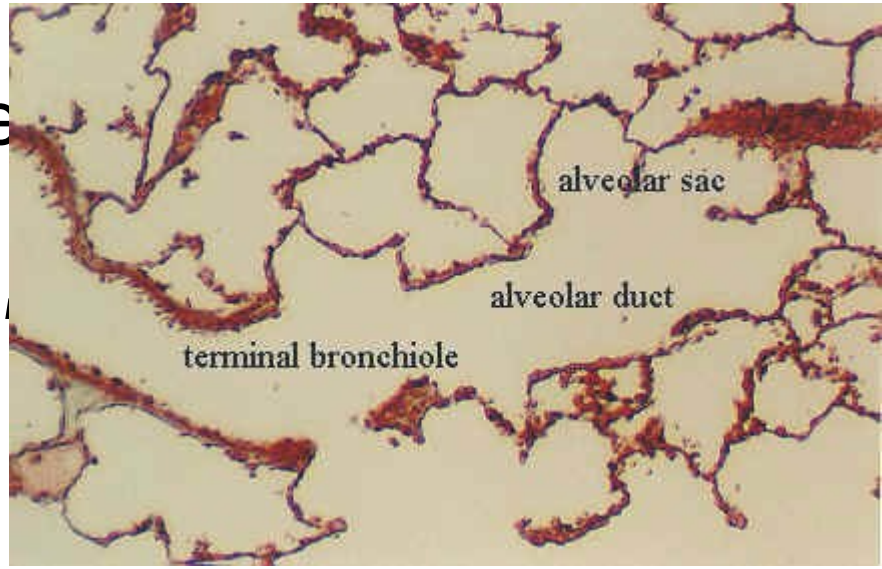
. **Lung: Bronchiole and Terminal Bronchiole**

Human • H.E. stain • Low magnification

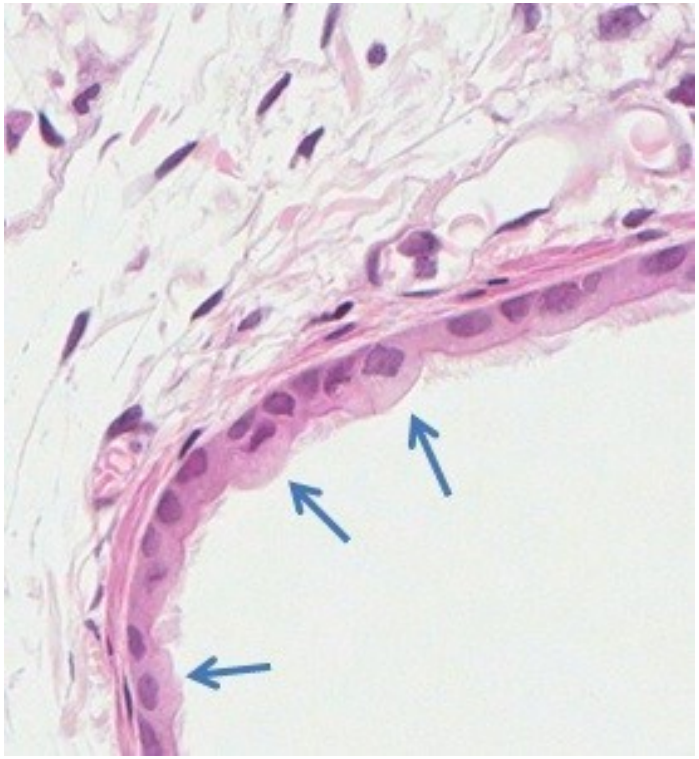


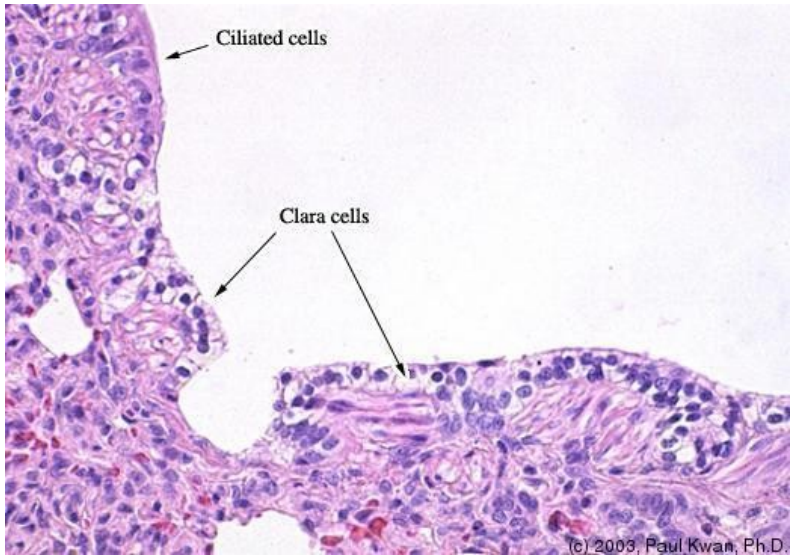
- **Terminal bronchioles**
- 0.5mm or less in diameter
- Mucosa Is lined by a single layer of cuboidal ciliated cells among which are scattered some non ciliated cells, commonly called **Clara cells** .
- Mucosa does not contain any glands.
- Terminal bronchioles divide into short, branching tubes the respiratory bronchioles,

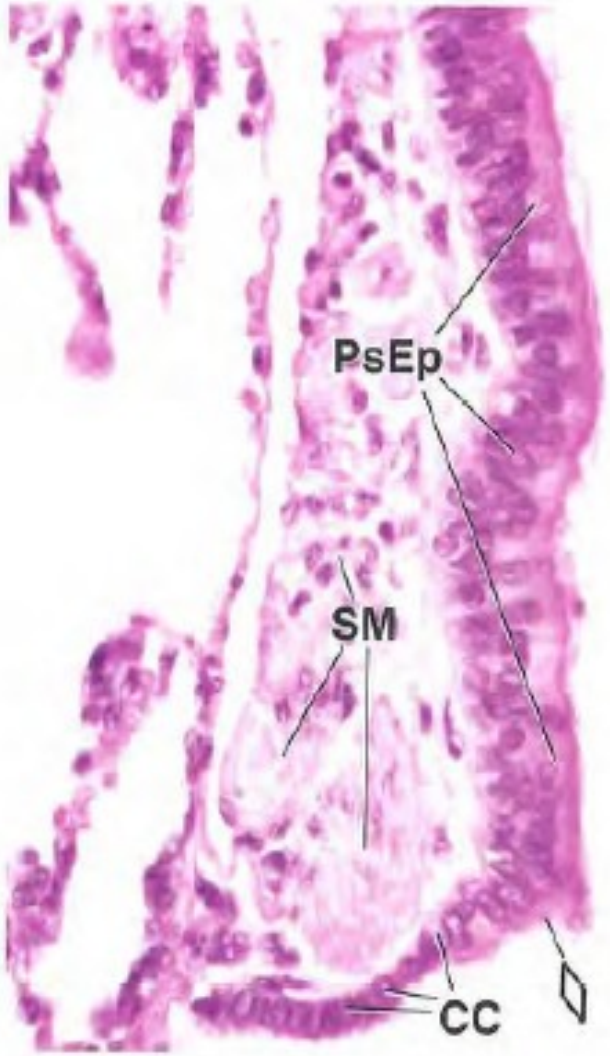




- **Clara cells** are dome-shaped cells.
- These mitotically active clara cells secrete one of the component of surfactant.
- They act as stem cells
- Secrete proteins that protect the bronchiolar epithelium against oxidative pollutants.
- They contain enzyme systems which can detoxify noxious substance.
- The lamina propria is composed of smoot muscles & elastic fibers.

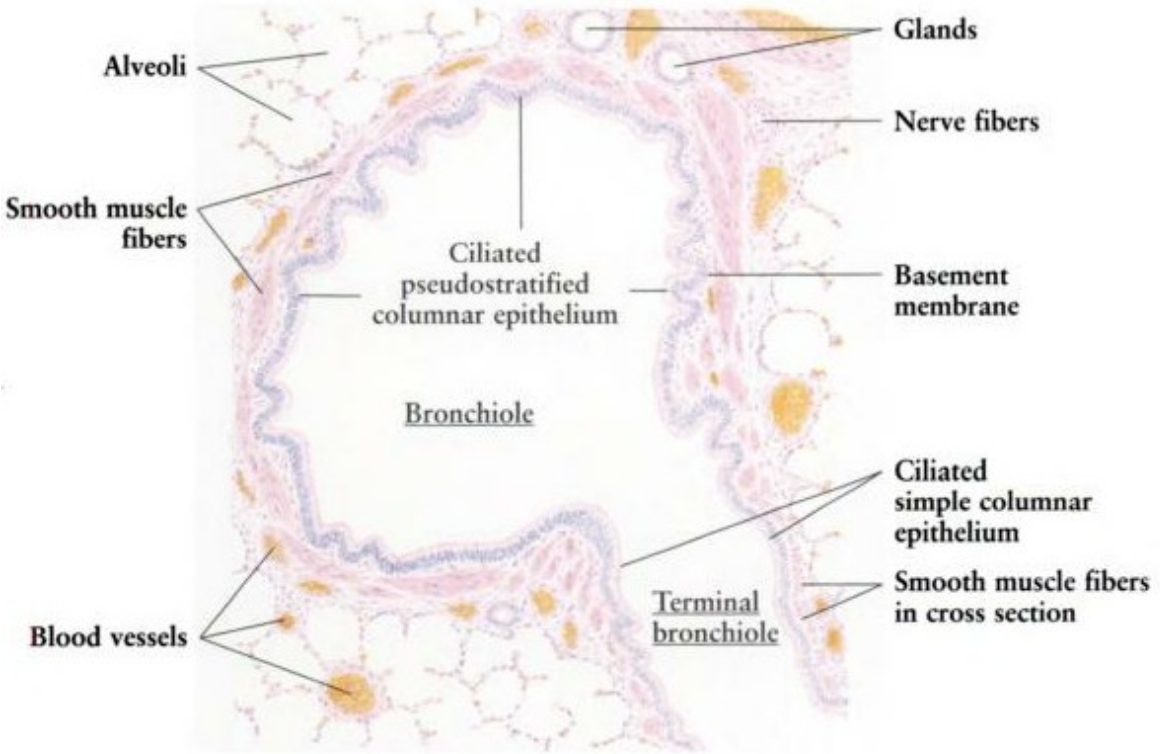






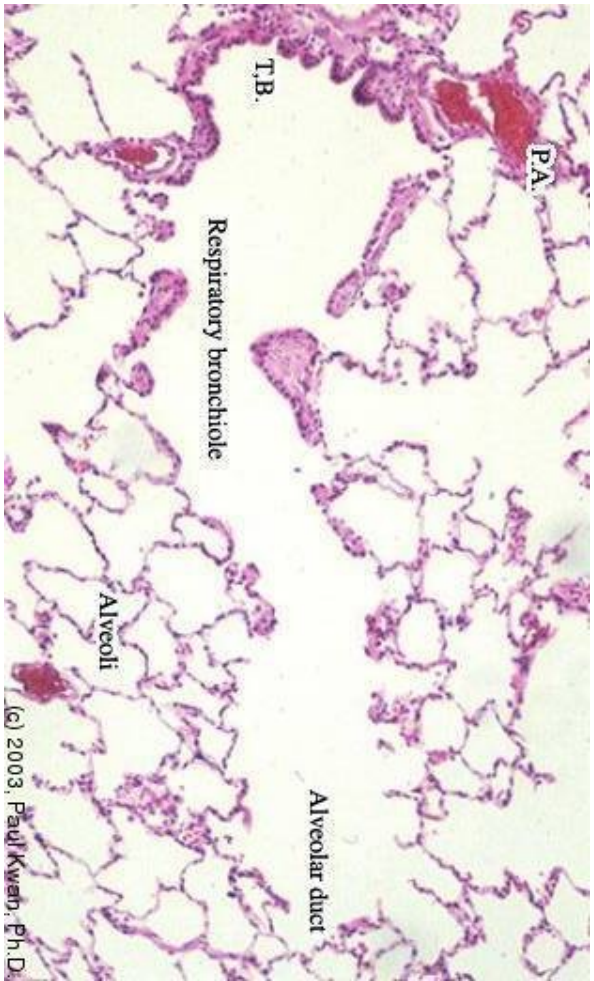
. **Lung: Bronchiole and Terminal Bronchiole**

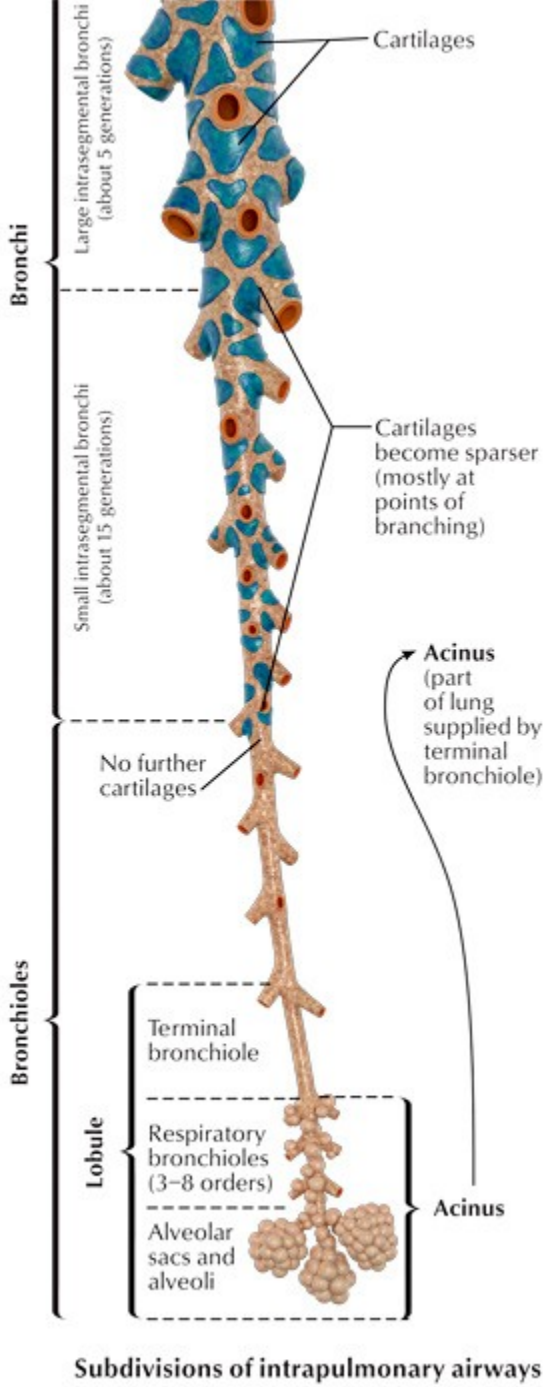
Human • H.E. stain • Low magnification



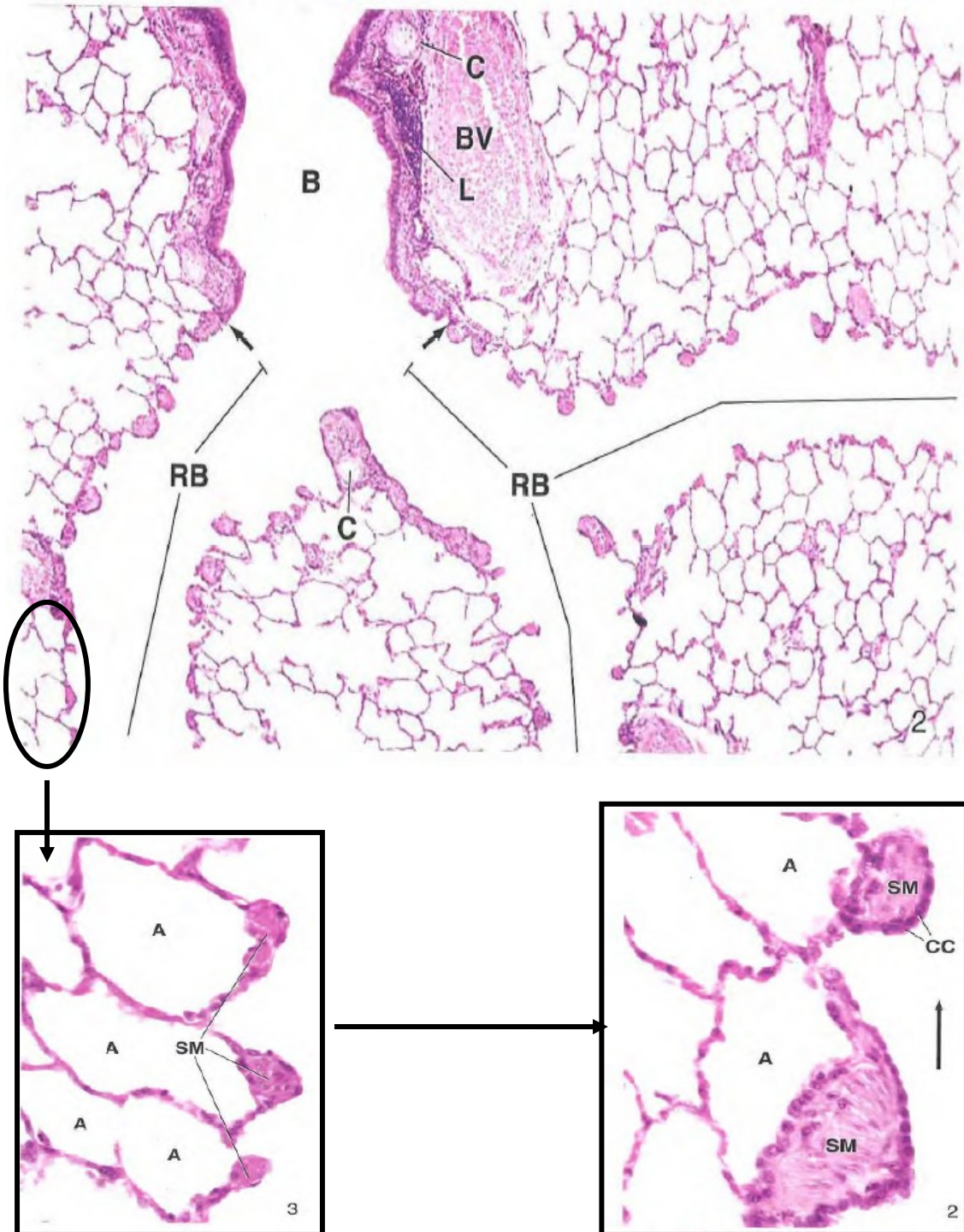
Respiratory bronchioles

- Are the first structures that belong to the respiratory portion of the respiratory system.

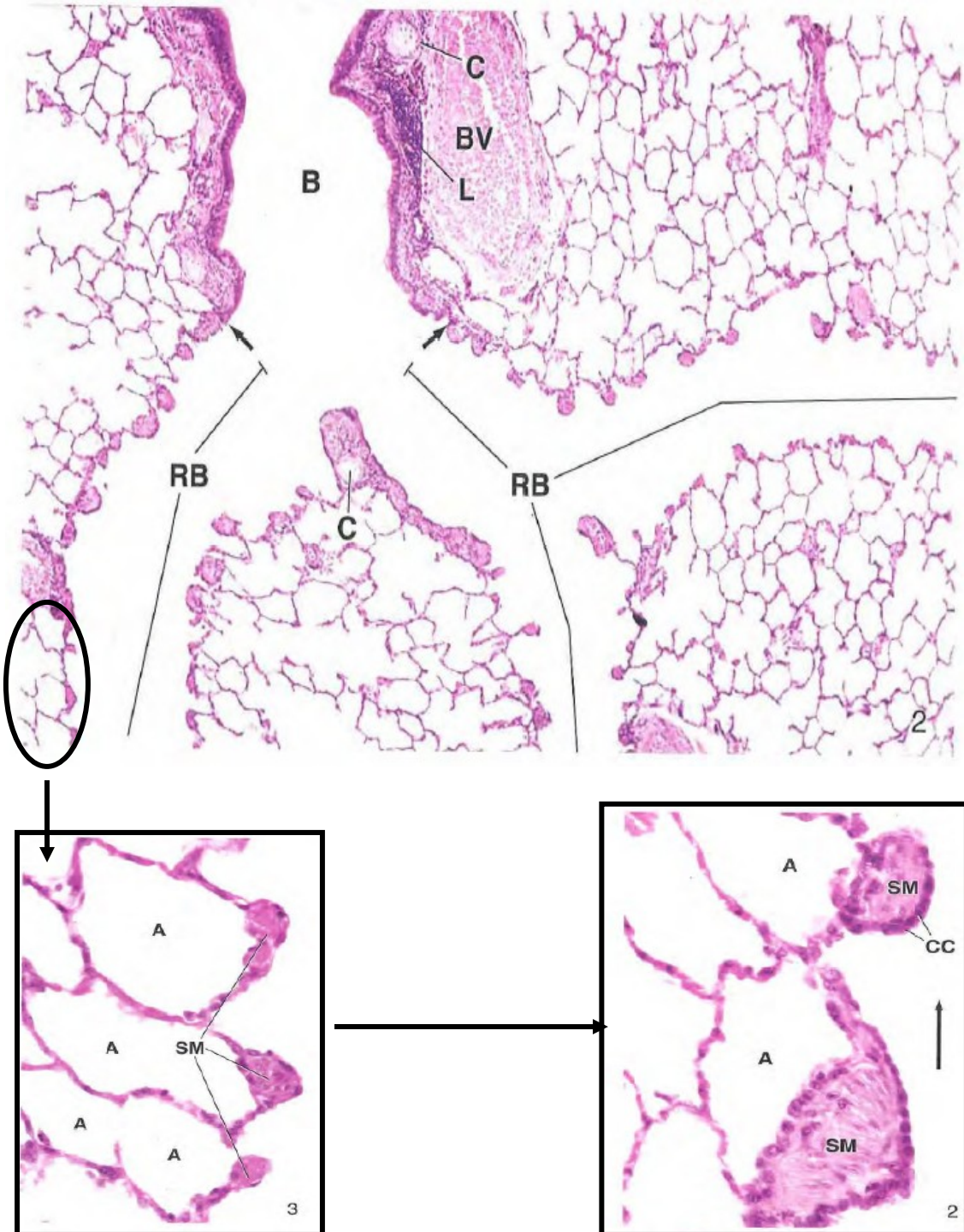




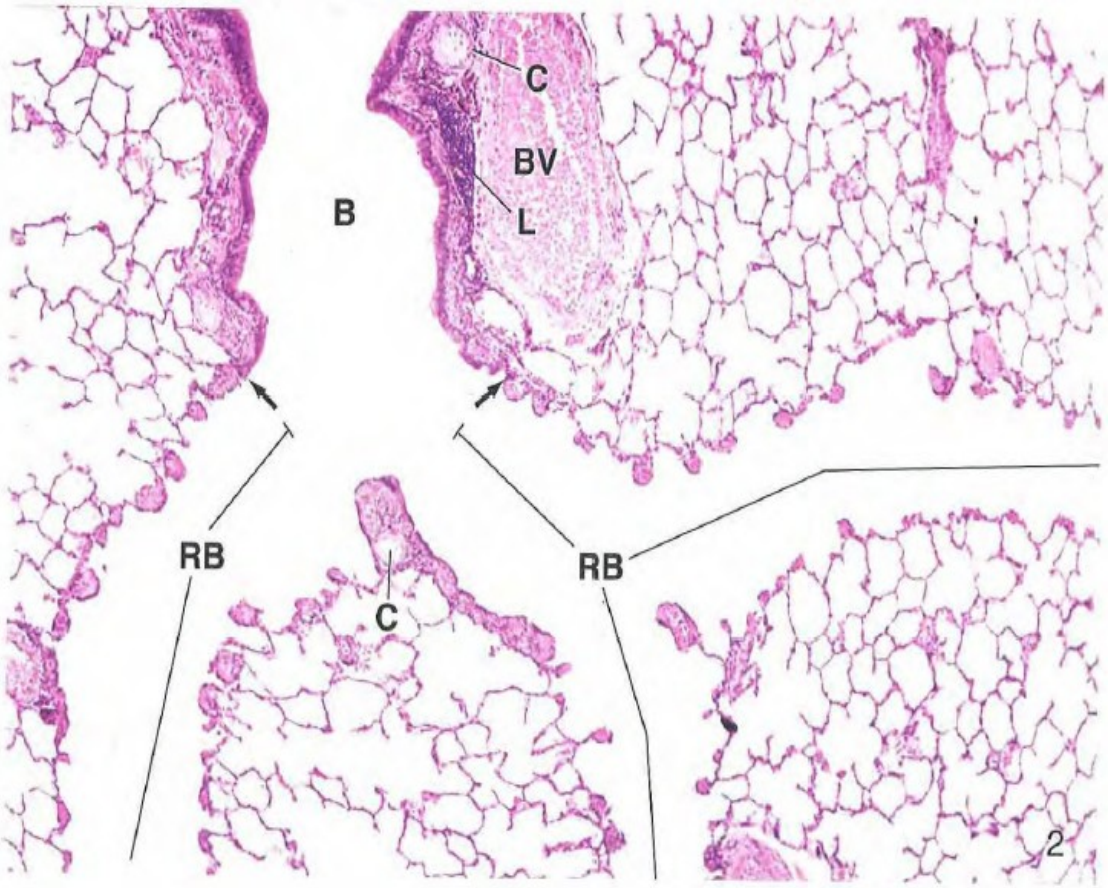
- The respiratory bronchiole has a wall composed of two components:
- **One component of respiratory bronchiolar wall** consists of recesses that have a wall similar to that of the alveoli.
- This component consists numerous openings along their length & has Small out-pocketing of the walls called alveoli, thus capable of gas exchange.
- The rim of alveolar opening the bronchiolar cuboidal epith becomes

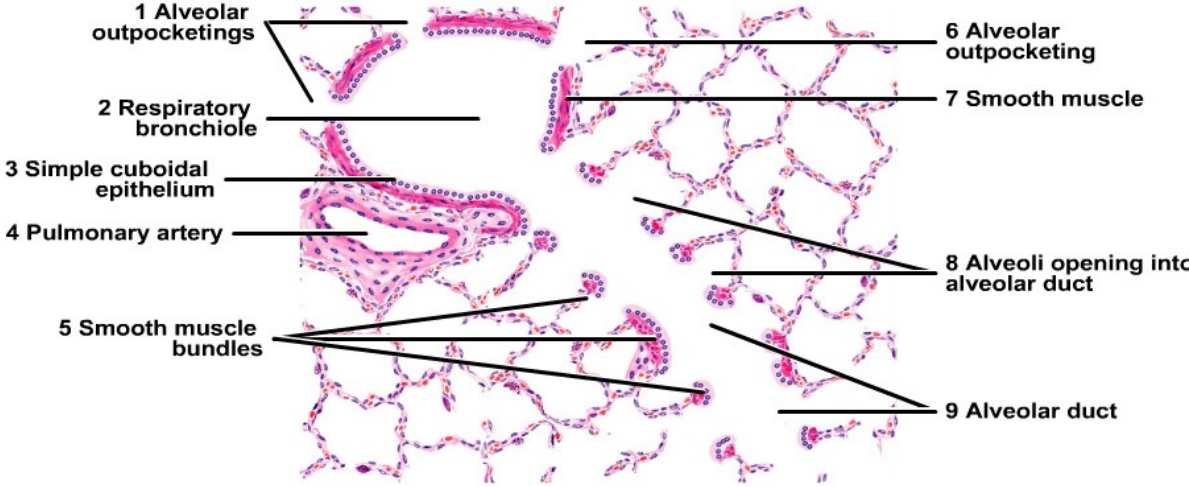


- The **other component of respiratory bronchiolar wall** is formed between the alveoli by small cuboidal cells that appear to rest on a small bundle of smooth muscle surrounded by a thin investment of connective tissue
- The larger respiratory bronchial wall is lined by simple cuboidal ciliated epithelium but becomes completely non-

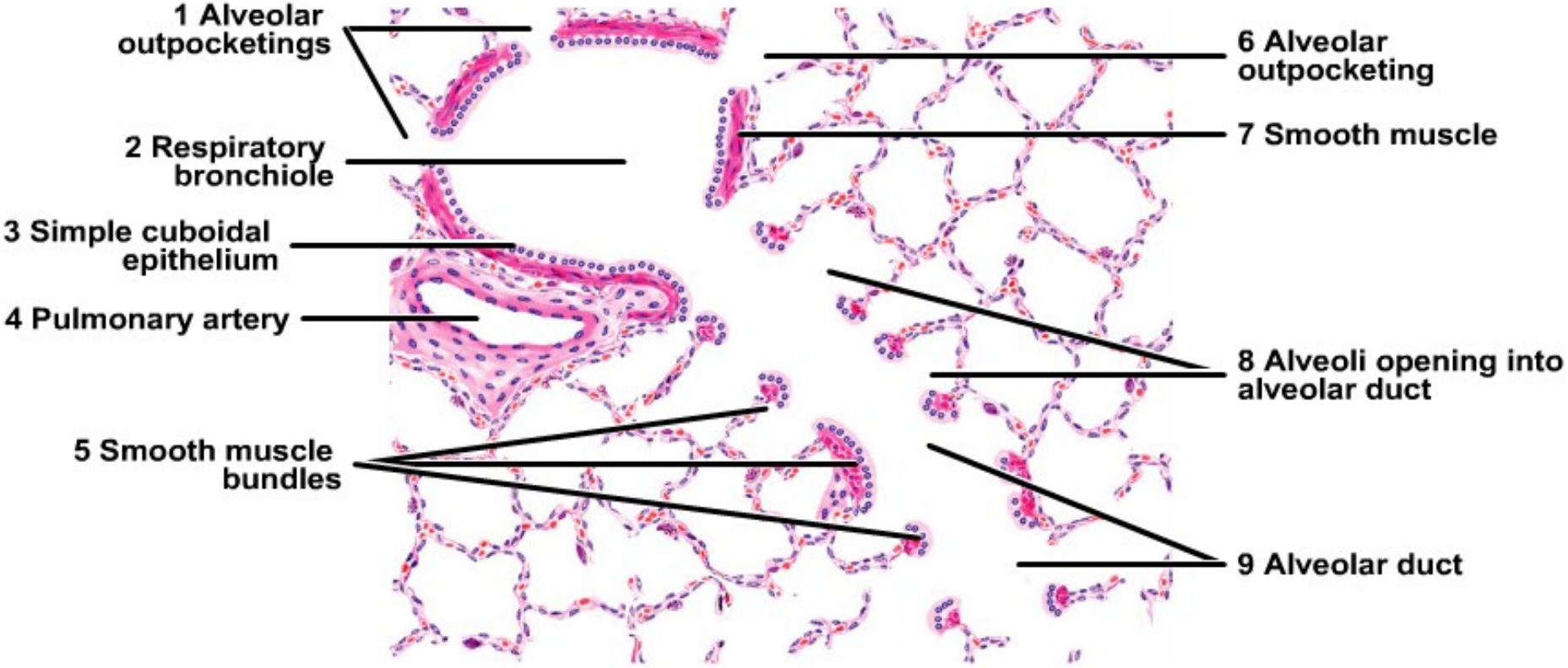


- The number of alveoli increases as the respiratory bronchioles continue to divide.
- Goblet cells & glands are absent.



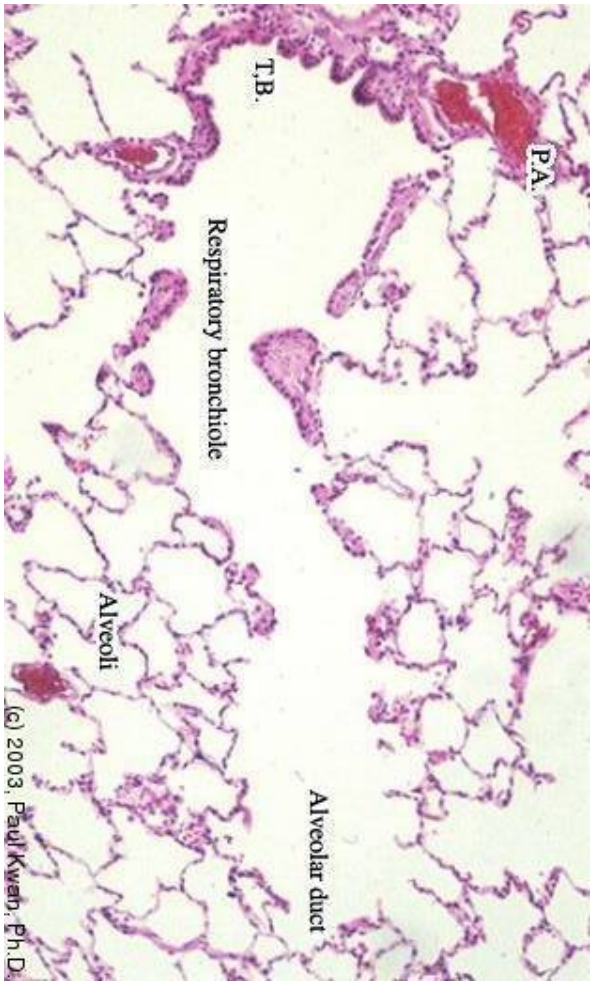


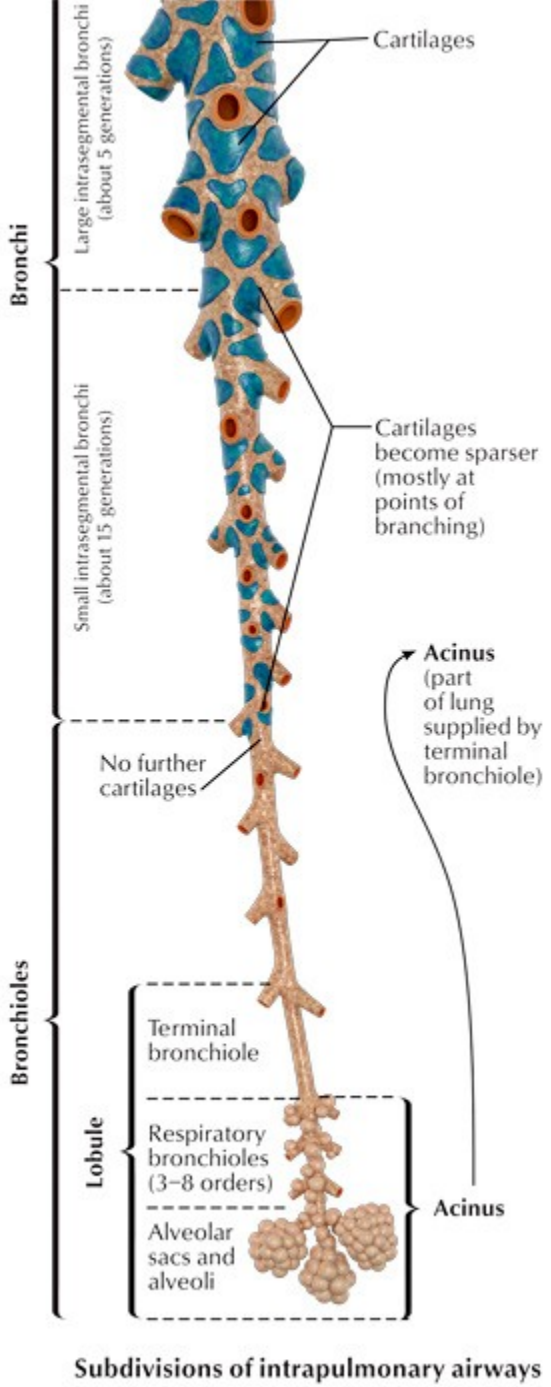
RESPIRATORY BRONCHIOLES



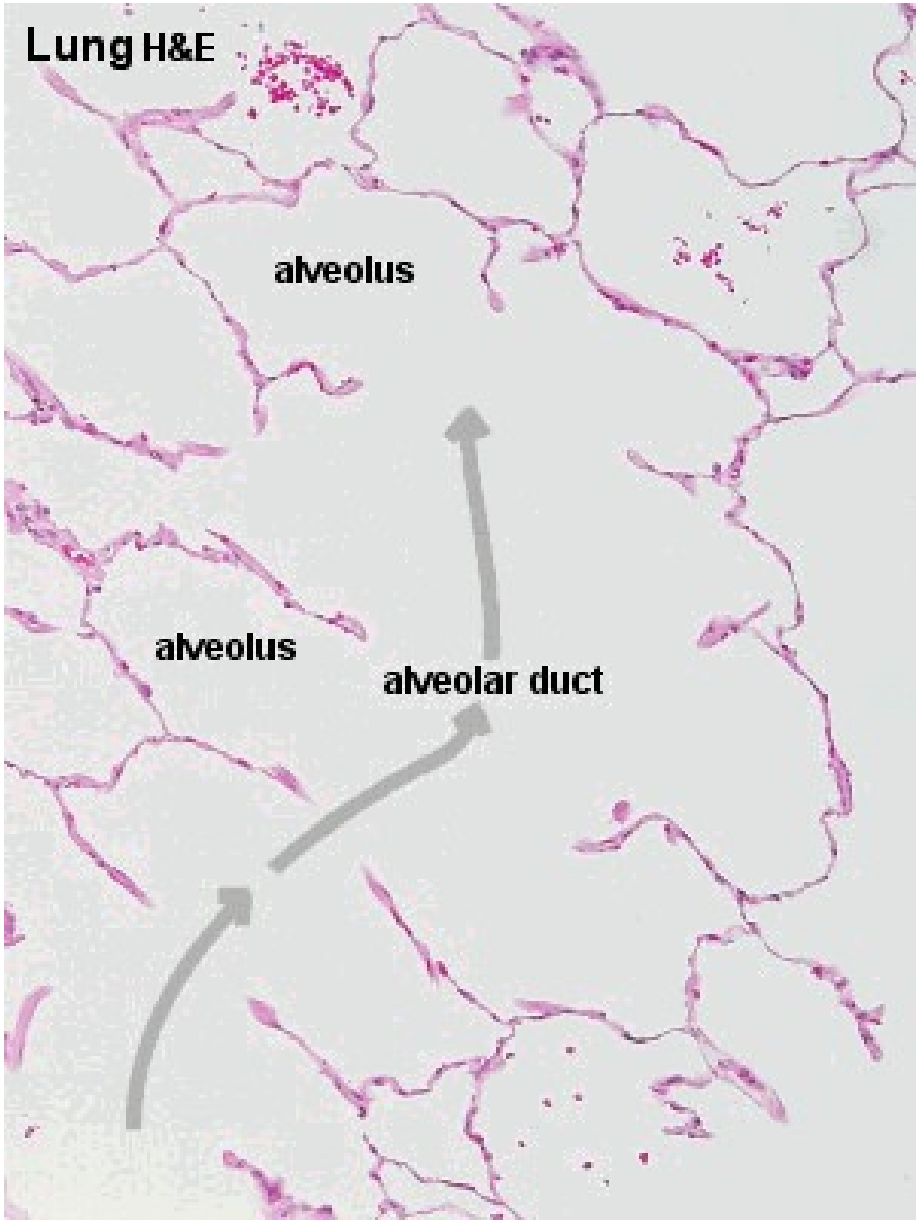
Alveolar ducts

- Preceding distally the number of alveoli increases as the respiratory bronchioles continue to divide & the tube is now called **Alveolar ducts**

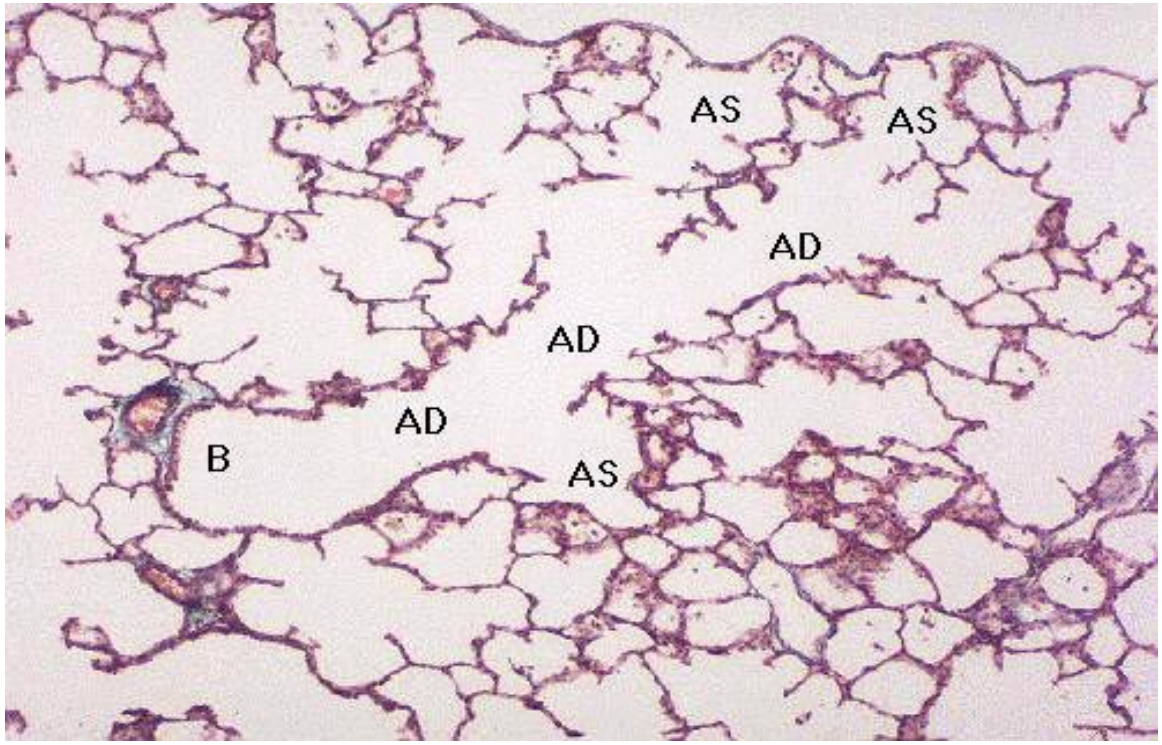


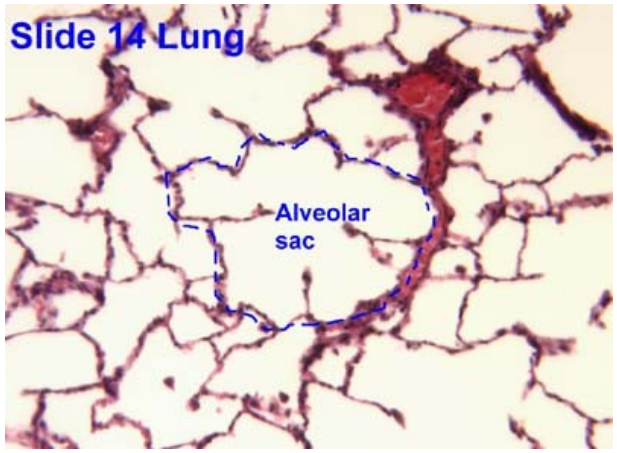


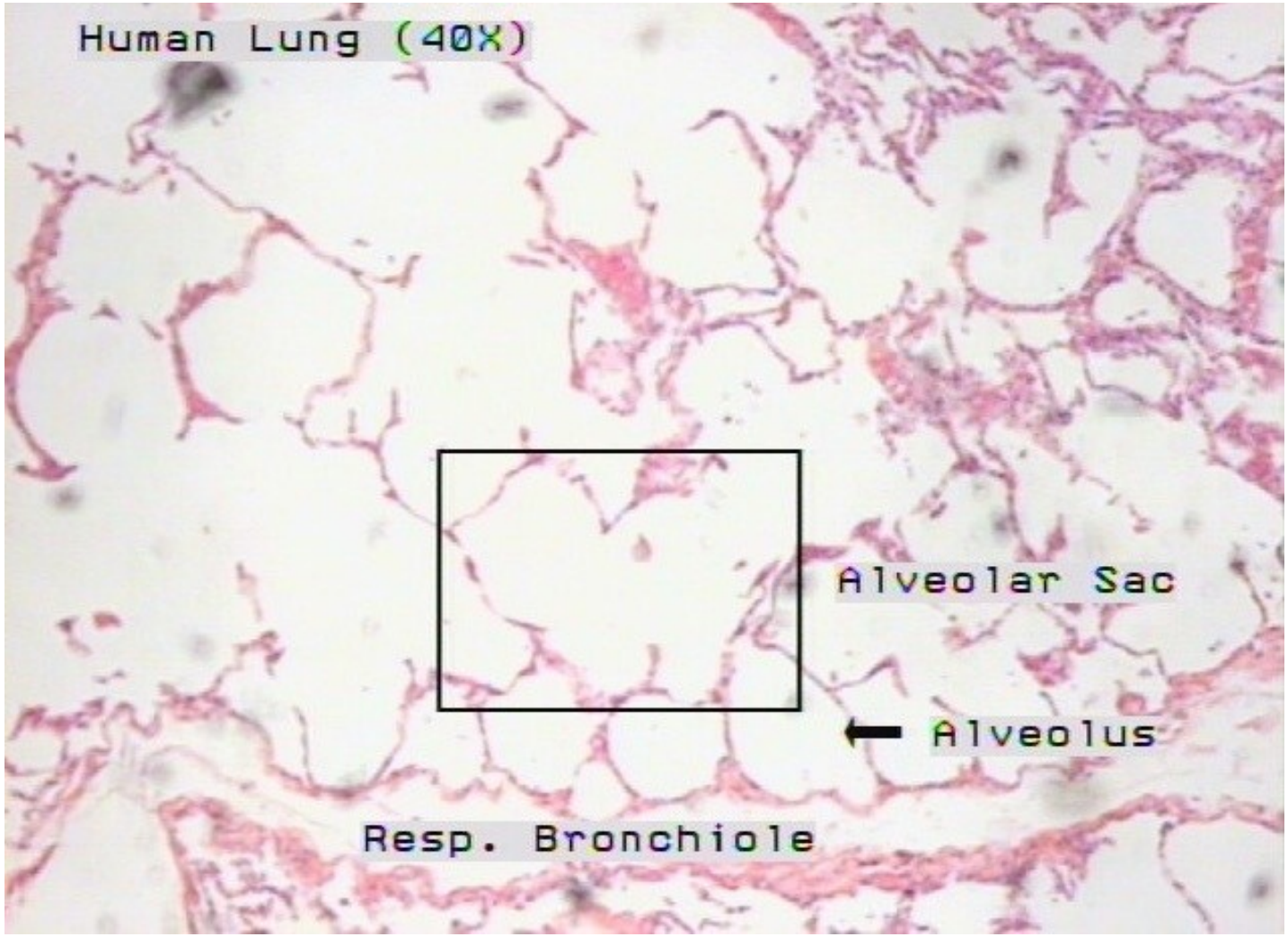
- The walls of alveolar ducts are so interrupted by alveoli openings along their length & has Small out-pocketing of the walls called alveoli and alveolar sacs (clusters of alveoli) the site of gas exchange.



- The alveolar ducts end in pulmonary atria which are the common chambers that give rise to two or more alveolar sac at the distal end.
- Groups of alveoli clustered together and sharing a common opening are referred to as an *alveolar sac*

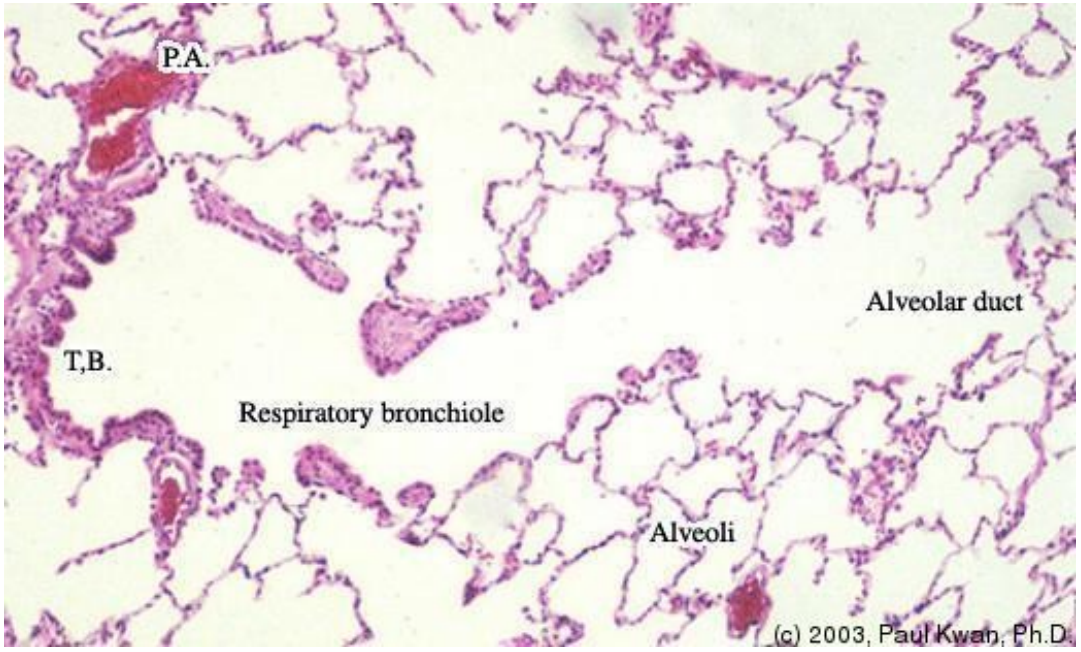






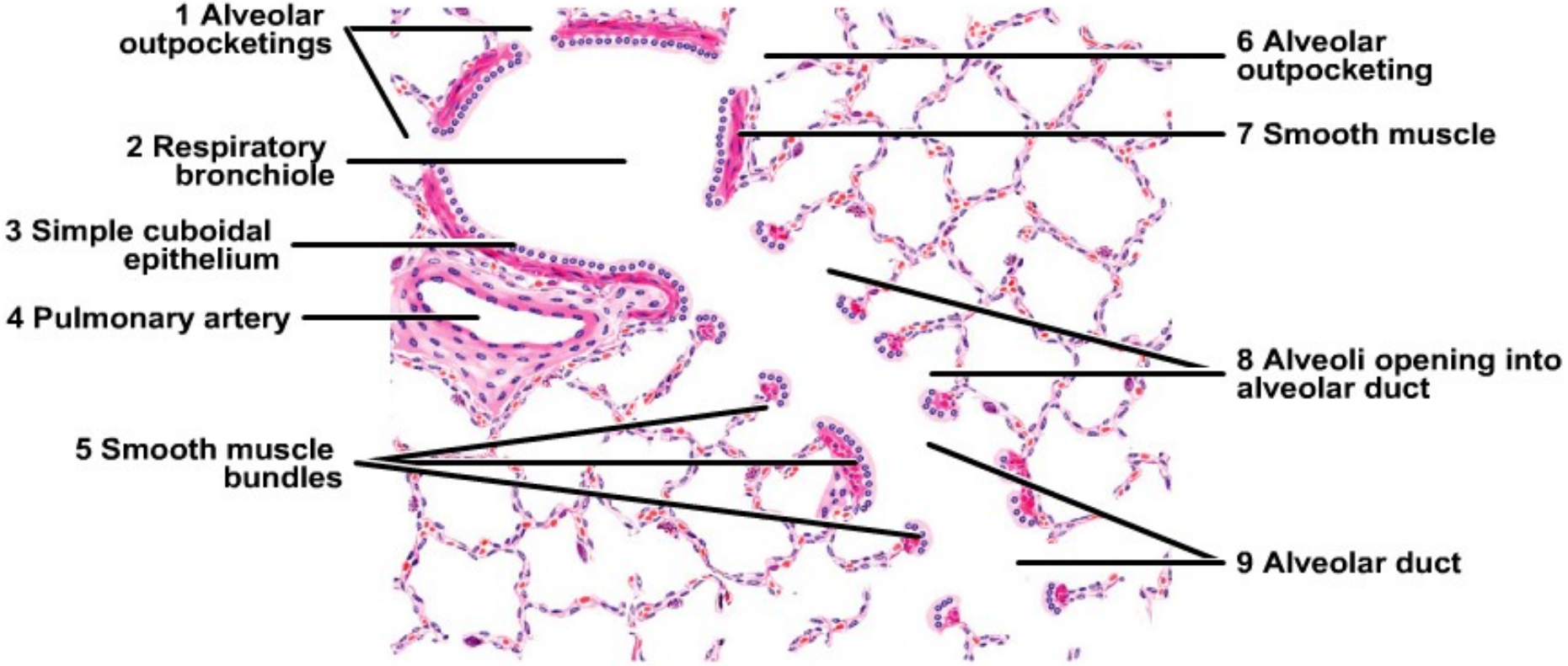
This is a low-power view of lung parenchyma. Shown in the rectangular area is an alveolar sac, composed of multiple individual alveoli, where the exchange of oxygen and carbon dioxide occurs. Each alveolus is lined by simple squamous epithelium.

- Both the alveolar duct & alveoli are lined by squamous alveolar cells.
- Smooth muscle surrounded by a thin investment of connective tissue
- Smooth muscle disappear at the end of alveolar ducts.



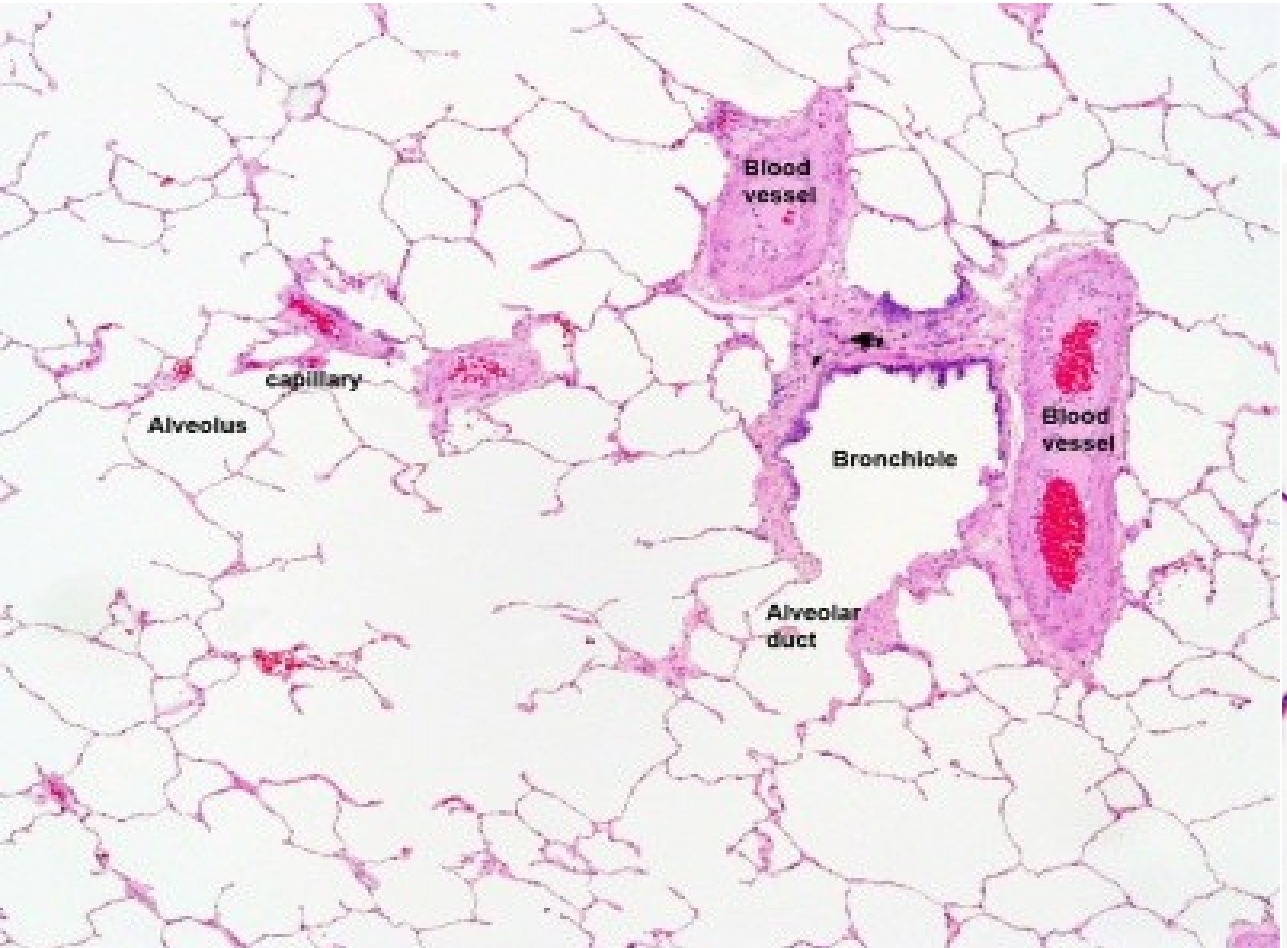
bronchioles (B), alveolar ducts (AD), alveolar sacs (AS)

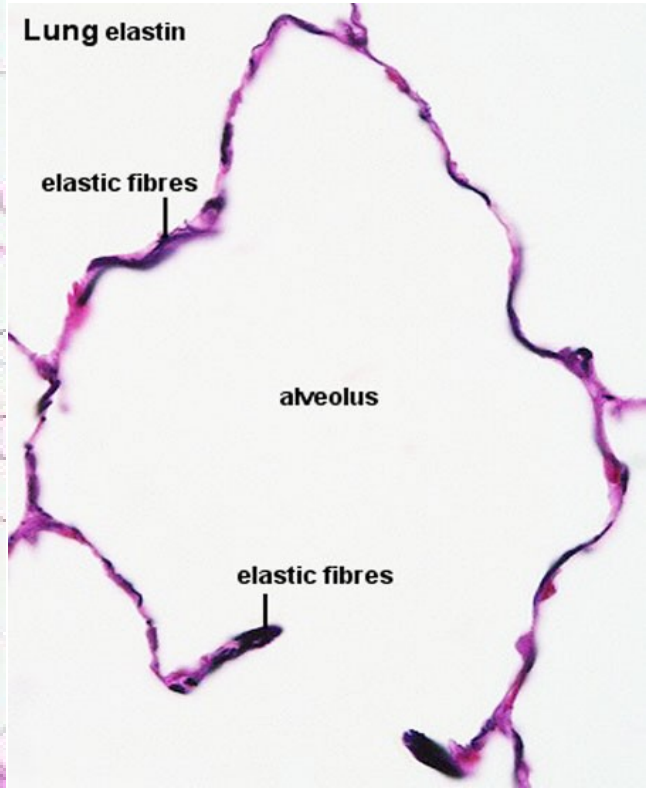
Alveolar duct and alveolar sacs



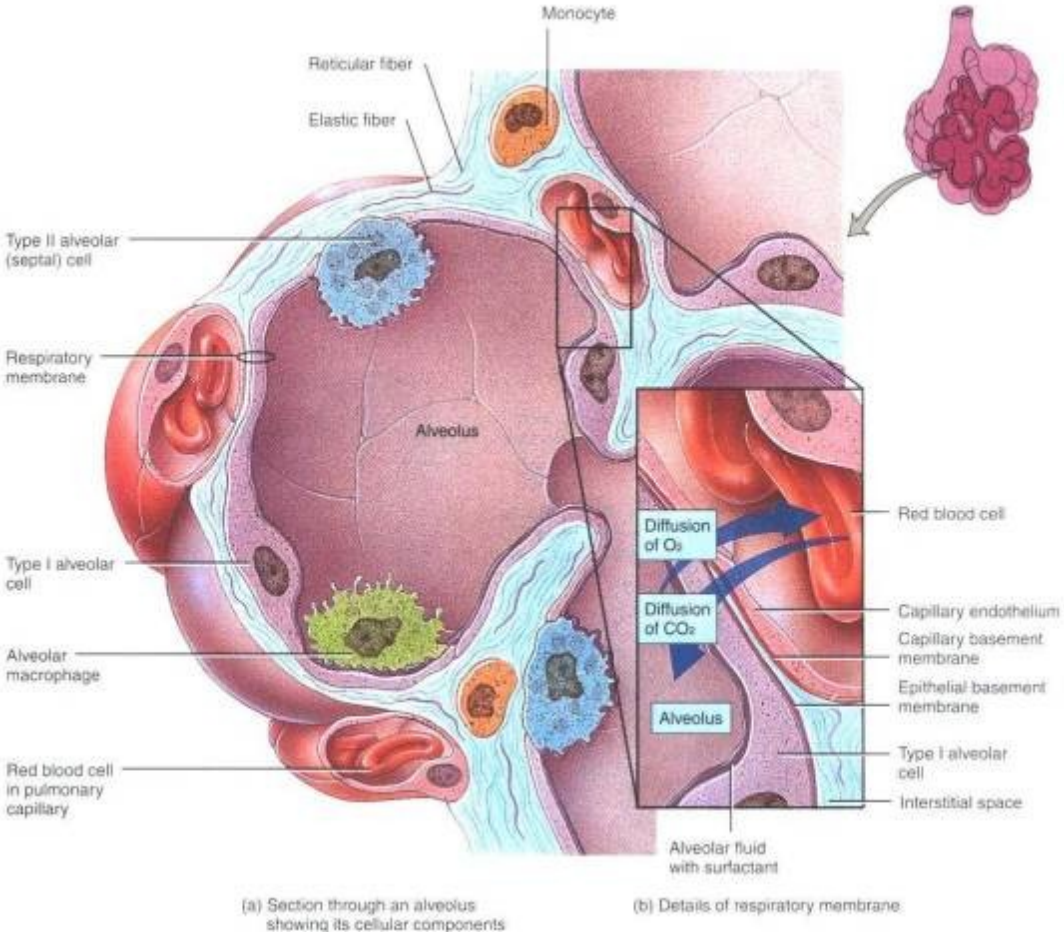
Alveoli

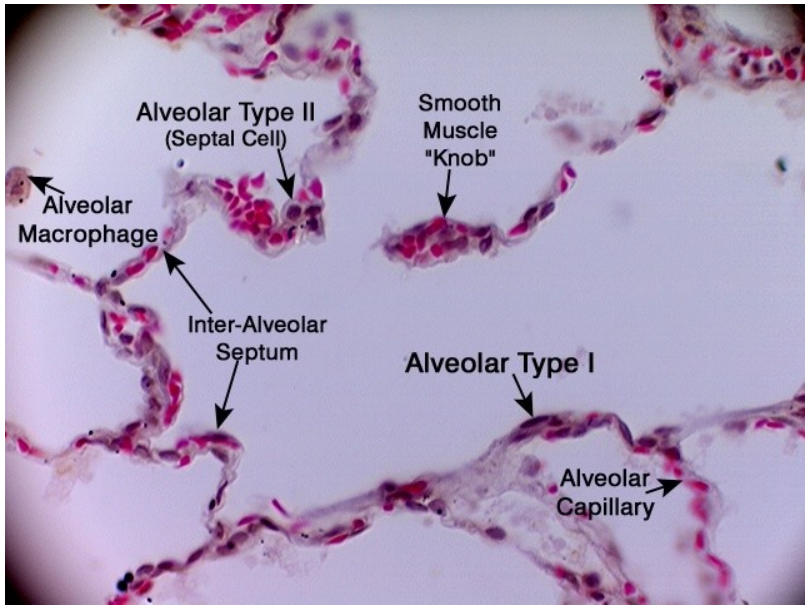
- Thin walled Cup shaped structure
- 0.2mm in diameter
- Opens either into respiratory bronchioles or alveolar ducts or alveolar sac.



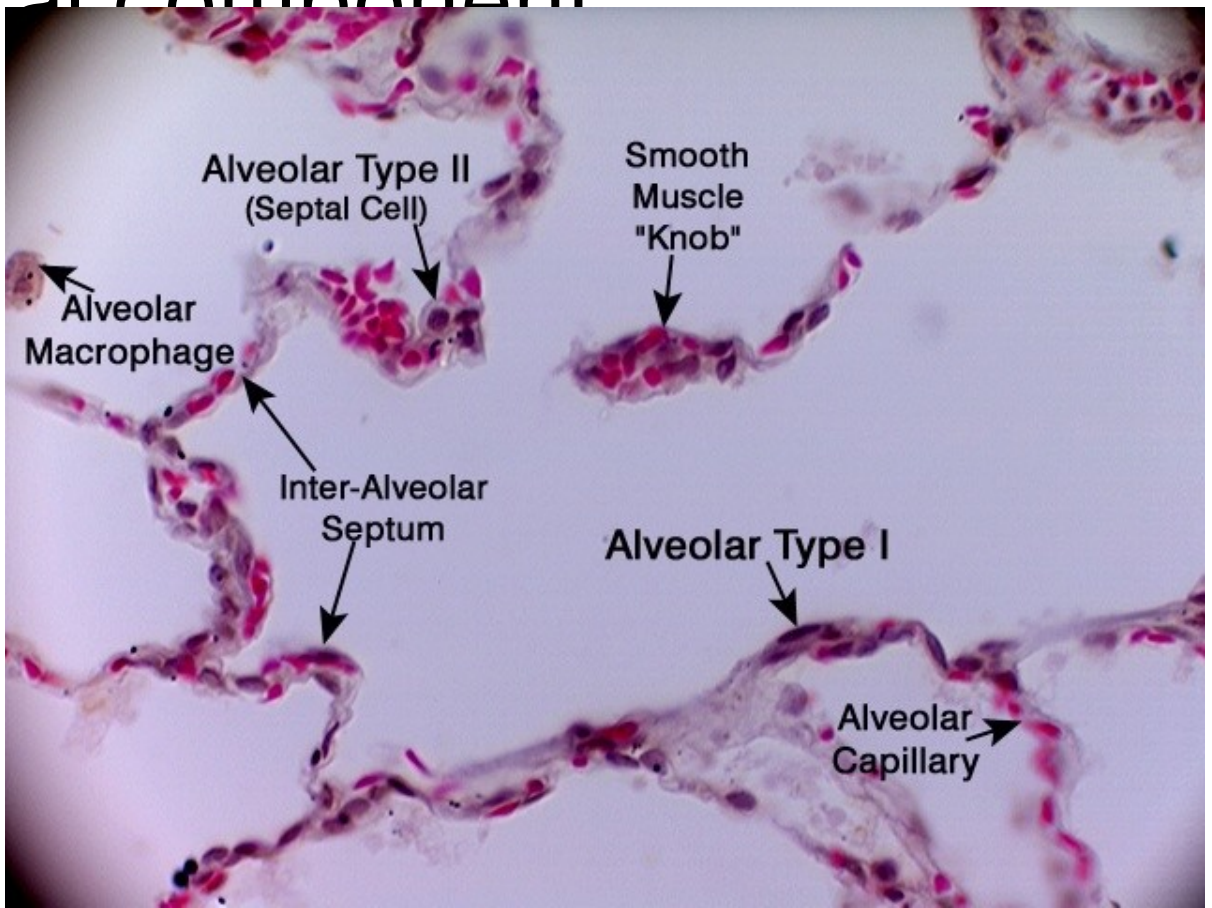


Each alveoli does not have complete separate wall. Rather, adjacent alveoli are separated from each other by *interalveolar septa*.



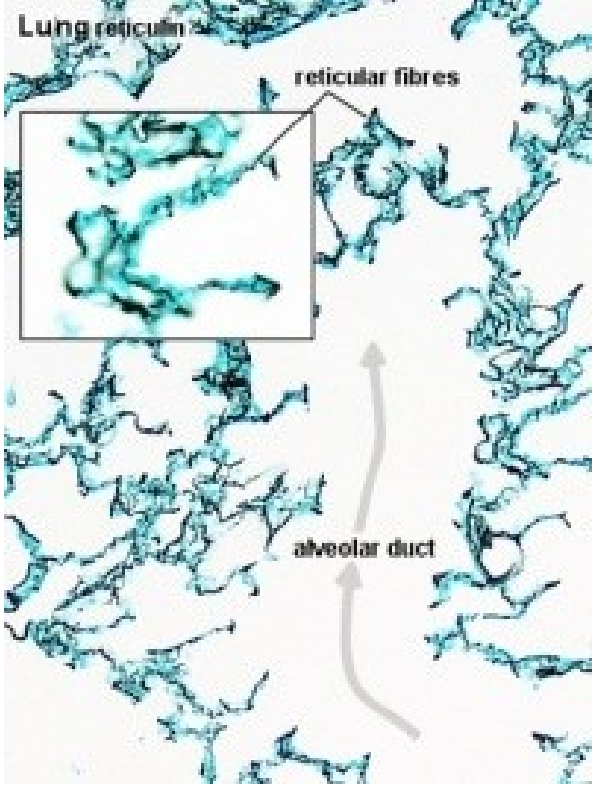


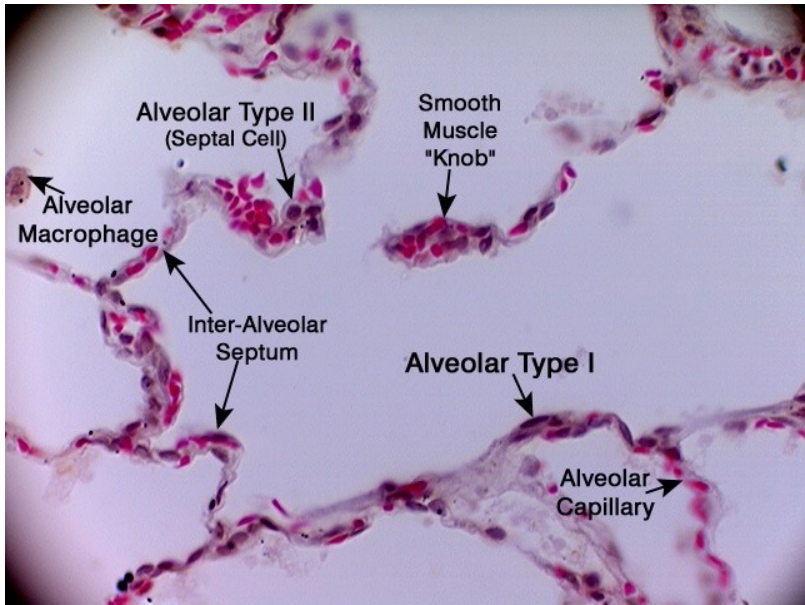
- **Histological Structure of Alveoli**
- The wall of the alveoli is formed by a thin sheet (~2µm) of tissue separating two neighboring alveoli. This sheet is formed by epithelial cells on each side and central component



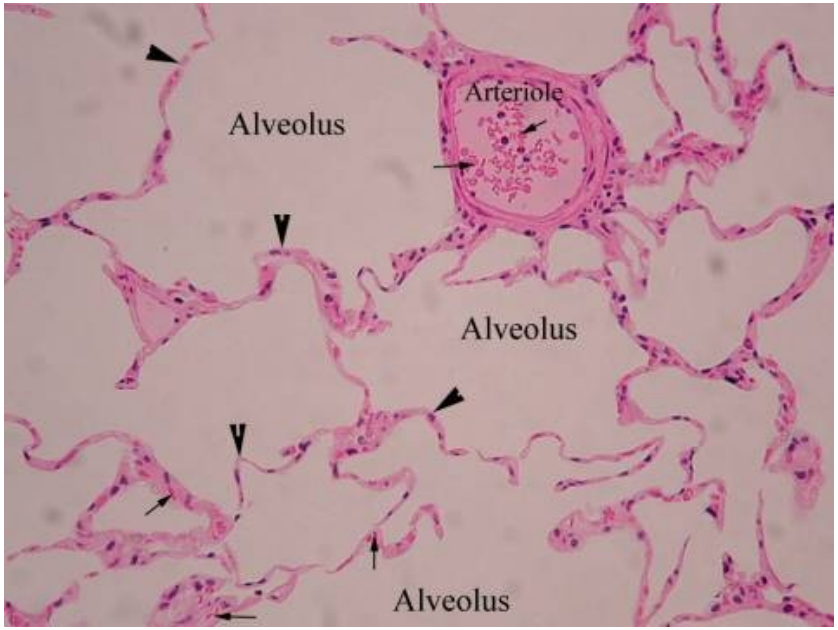
- **Histological Structure of Alveoli**

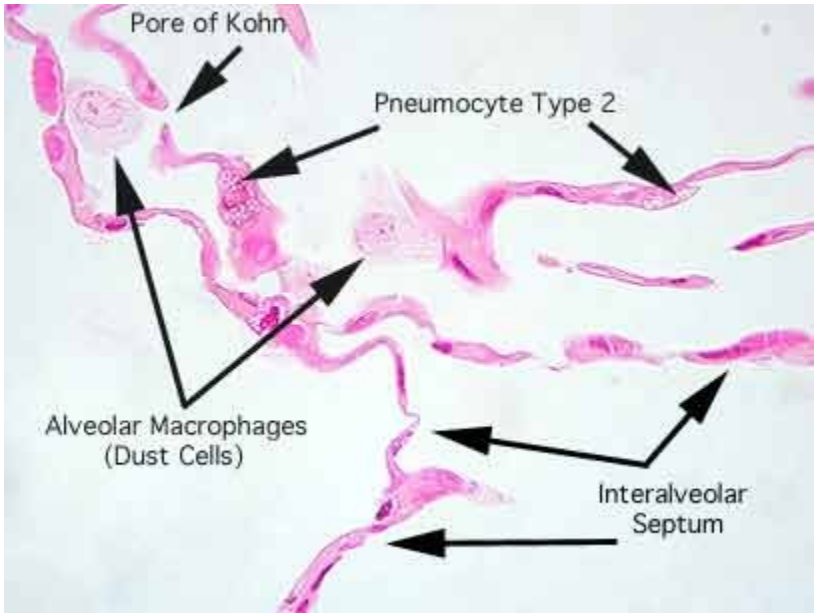
- The central component of the wall is the capillary and, in certain locations, associated Collagenous (few and fine), reticular and elastic fibres connective tissue are present.
- On each side, where it faces the alveolus, a flat squamous cell (pneumocyte type 1) is interposed between the capillary and the air spaces.



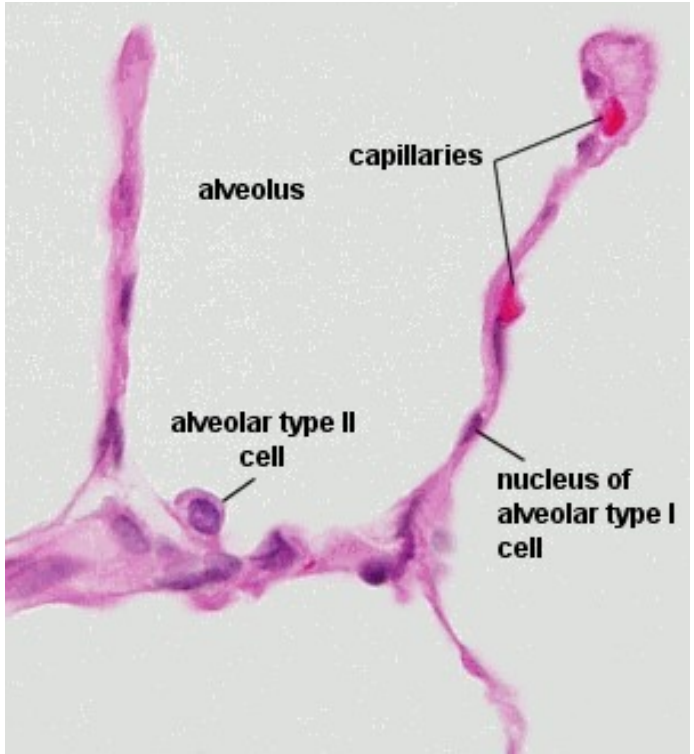


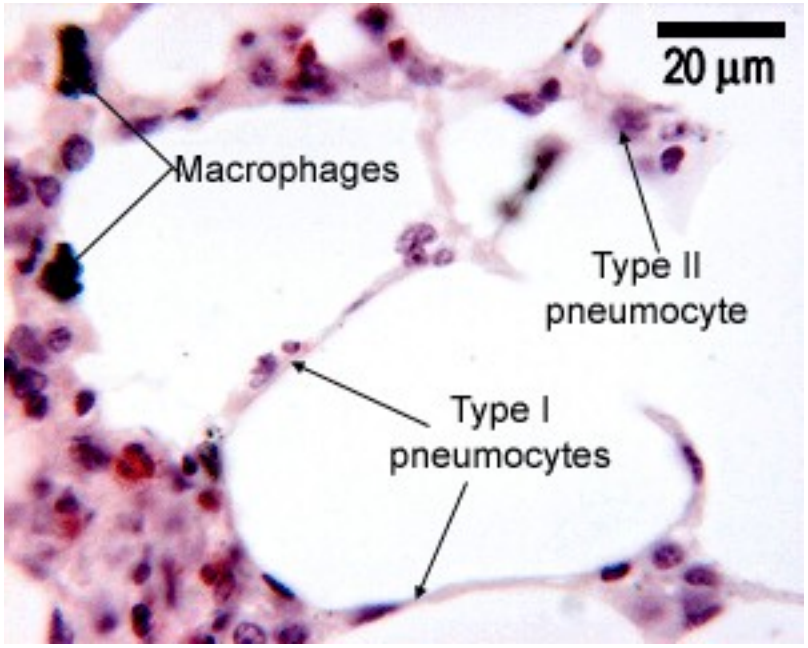
- Neighbouring alveoli may be connected to each other by small alveolar pores.



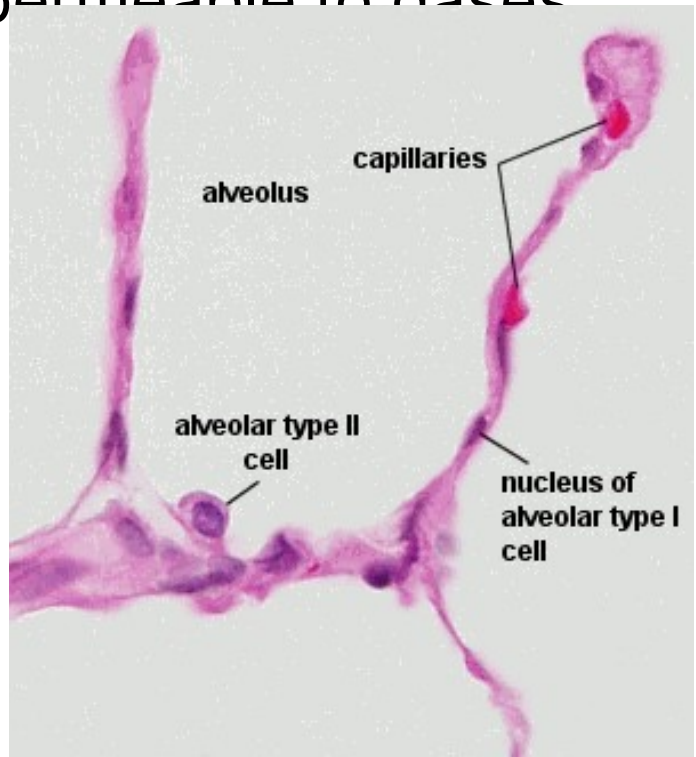


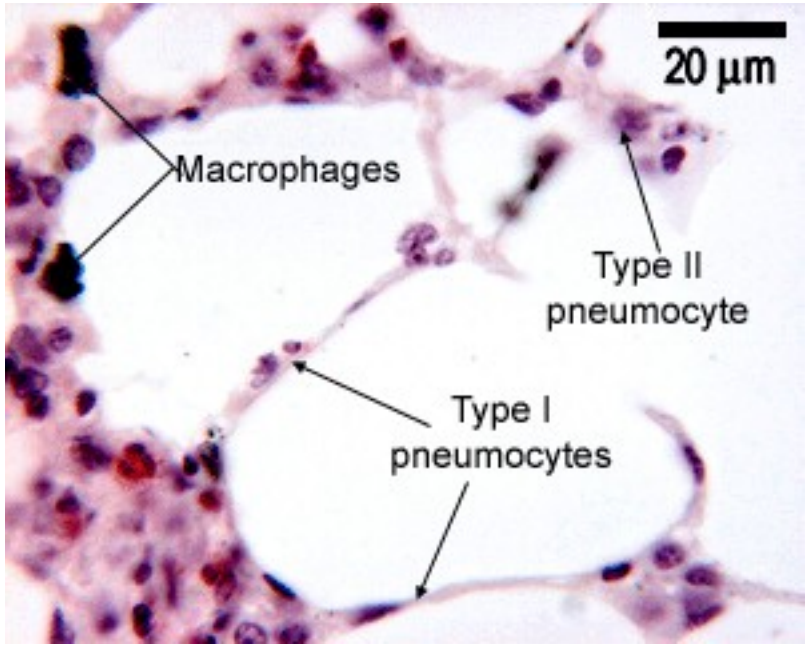
- The epithelium of the alveoli is formed by two cell types:
- **Alveolar type I cells**
- **Alveolar type II cells**





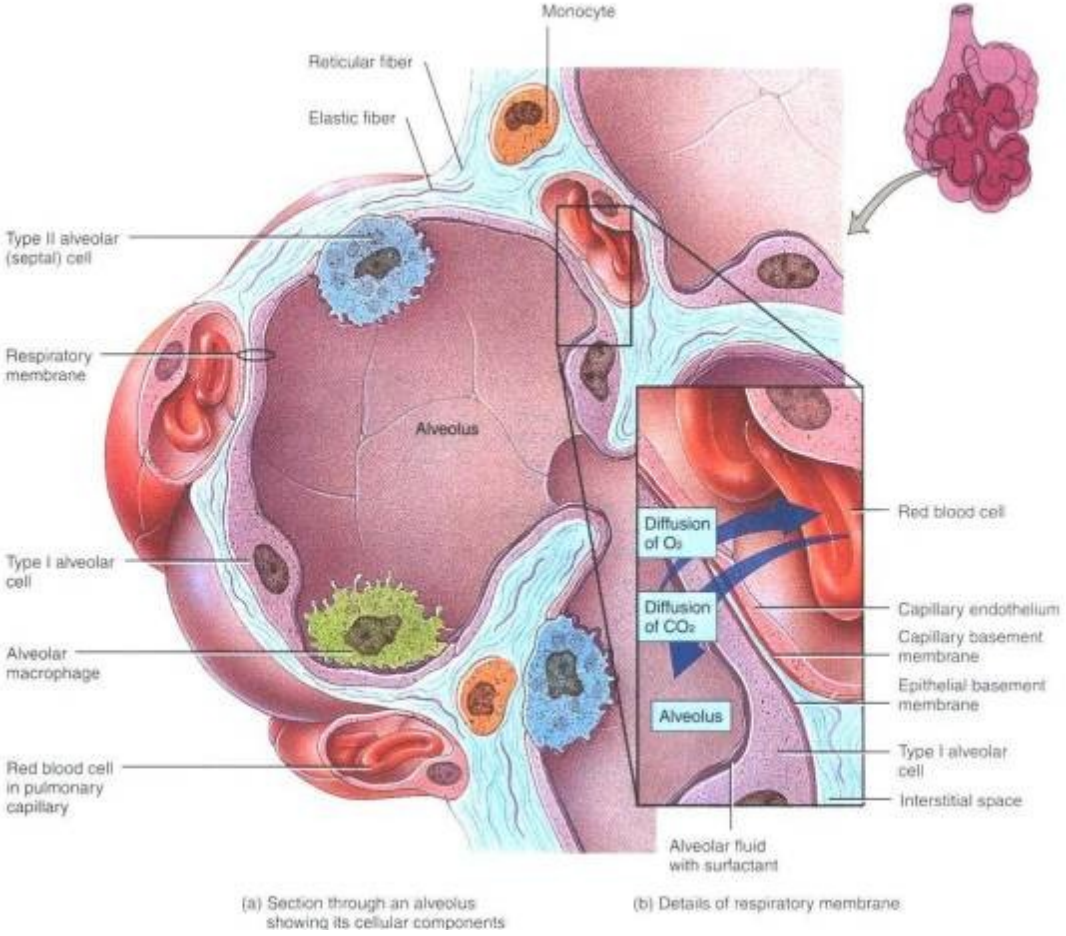
- **Alveolar type I cells** (small squamous alveolar cells or type I pneumocytes) are extremely flattened (thin as 0.05 µm or 25nm) and form the bulk (95%-97%) of the surface of the alveolar walls. The cells may actually form part of the epithelium on both faces of the alveolar wall. They are connected to each other by tight junctions. Organelles are present at centre of cell around the nucleus. **Cytoplasm contains pinocytotic vesicle which play a role in turnover of surfactant** . Main role of these cells is to provide the **barrier of minimal thickness** that is readily permeable to gases

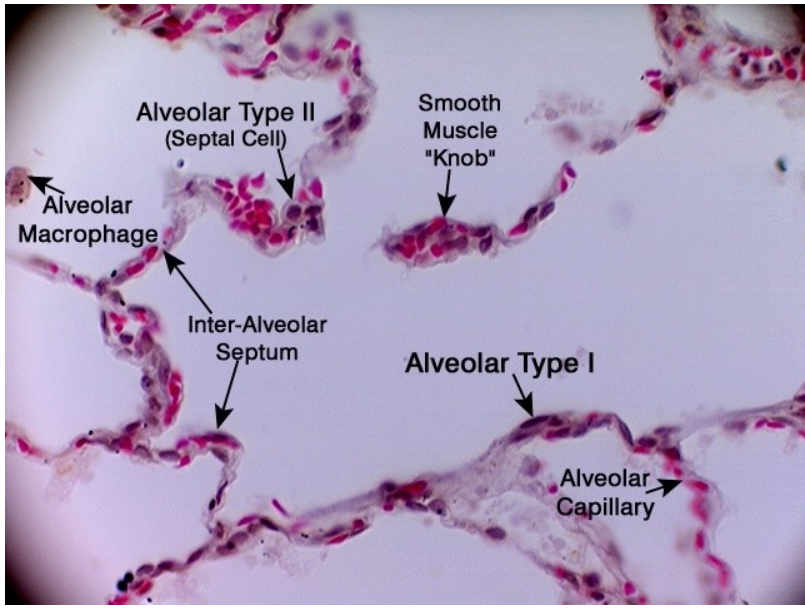




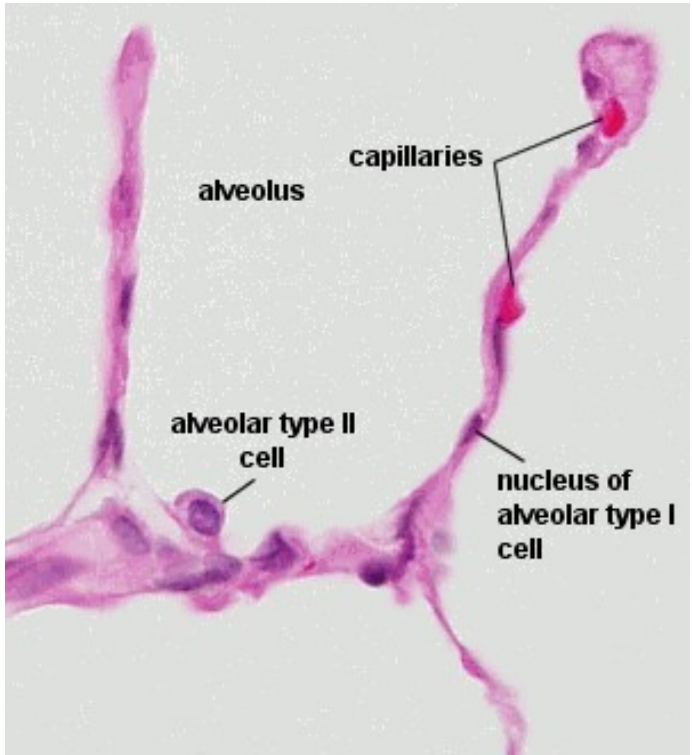

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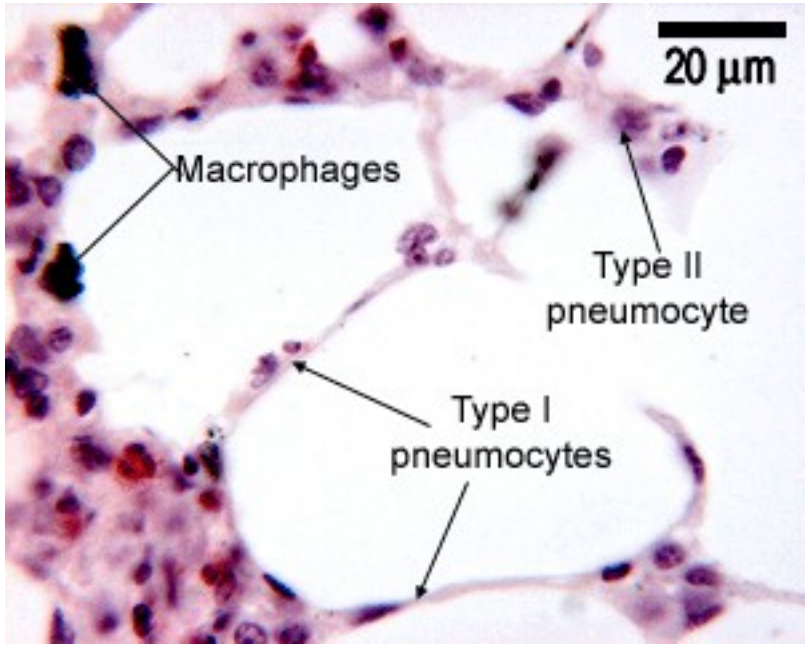
- In some places, the type 1 cell is separated from the capillary endothelial cell by a single basal lamina shared by the two cells. This is the thin portion of the alveolar-capillary complex. Gas exchange occurs through the thin portion of alveolar-capillary complex



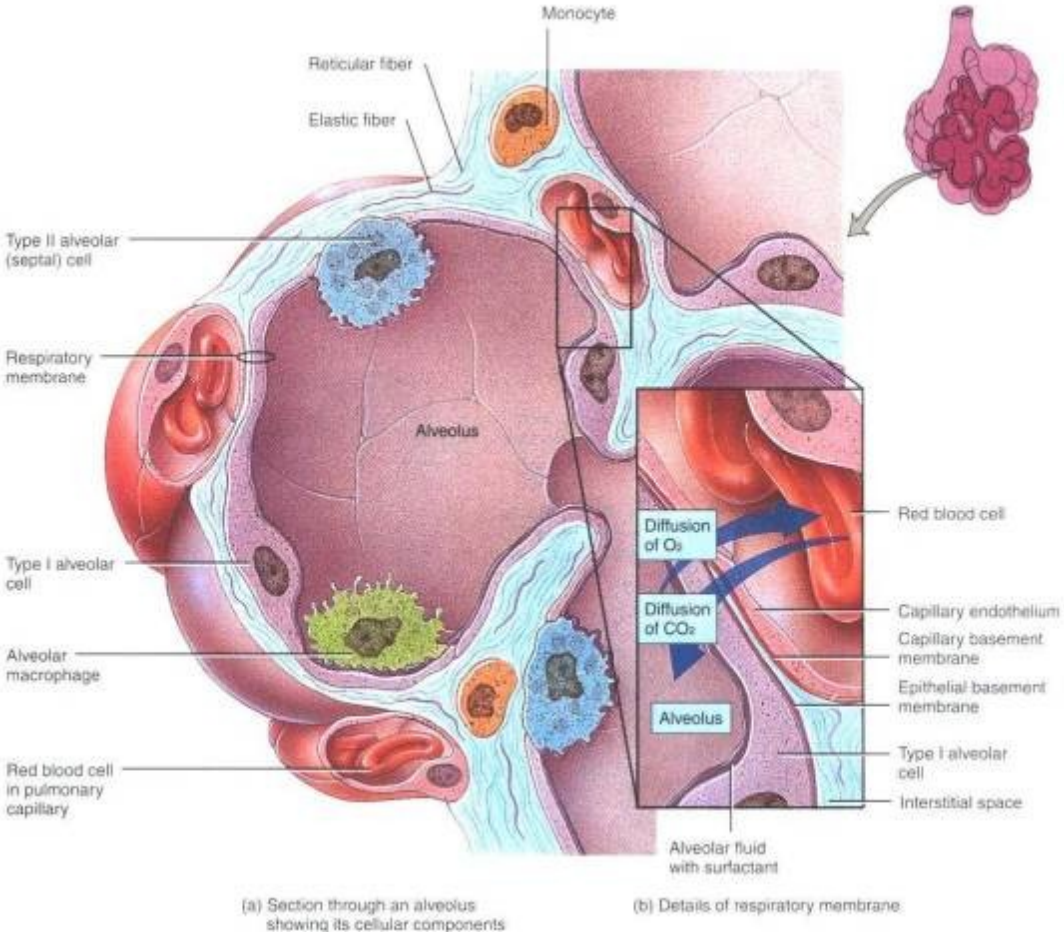


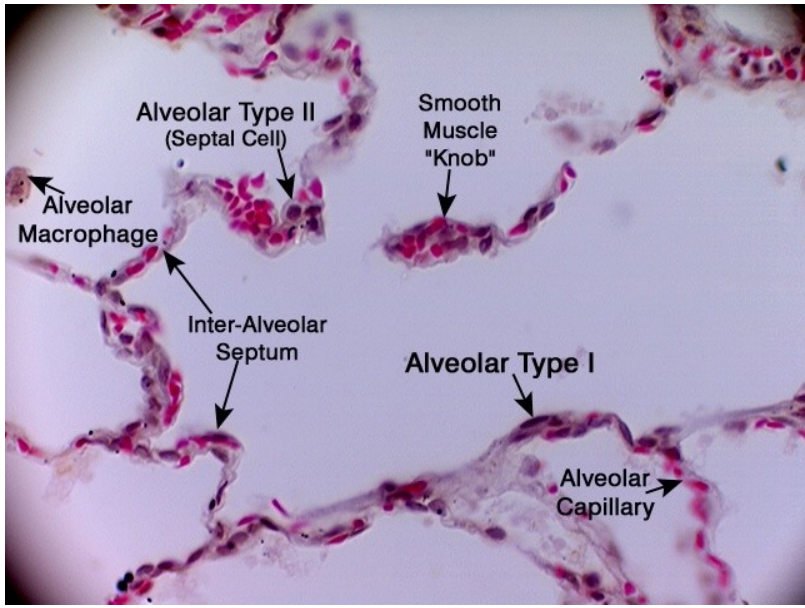
- **Alveolar type II cells** (the pneumocyte type II cell or septal cell) are rounded shaped. Form the (5%) of the surface of the alveolar walls. They rest on basement membrane. Nucleus is surrounded by a noticeable amount of cytoplasm which appears vacuolated . It also contain are large number of secretory granules, which consist of precursors to pulmonary surfactant made by the mixture of phospholipids, neutral lipids & four types of proteins A,B,C & D which keep surface tension in the alveoli low. They are also stem cells.





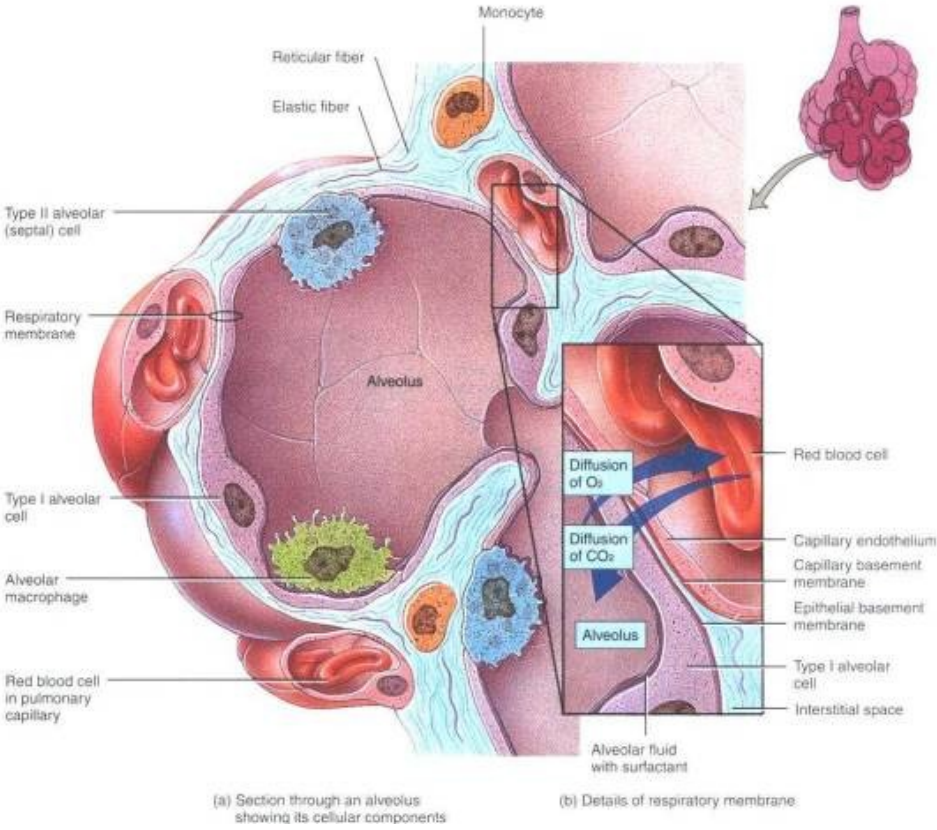
- **Alveolar type II cells** (the pneumocyte type II cell or septal cell)

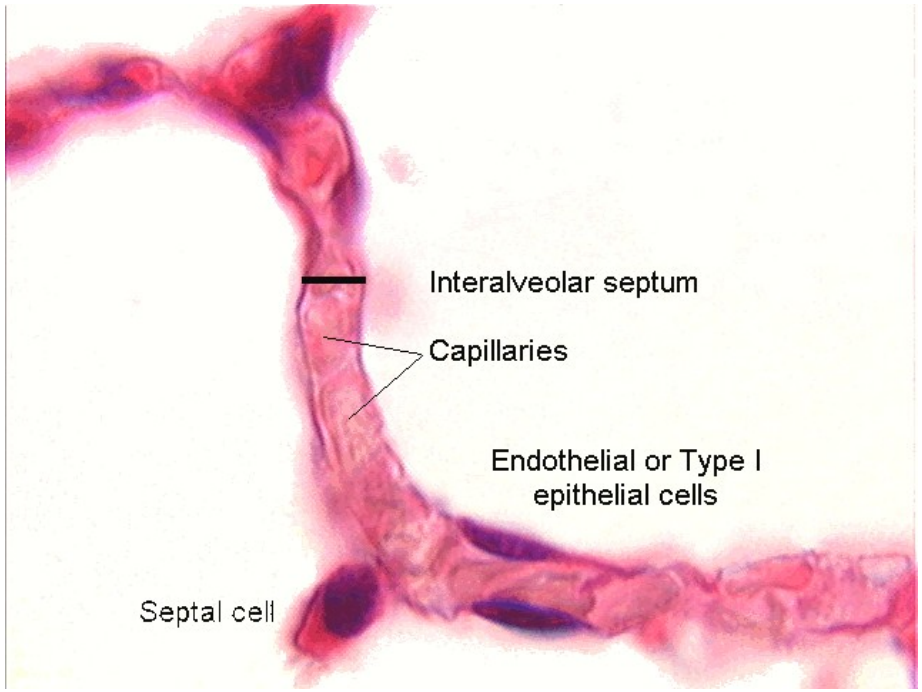




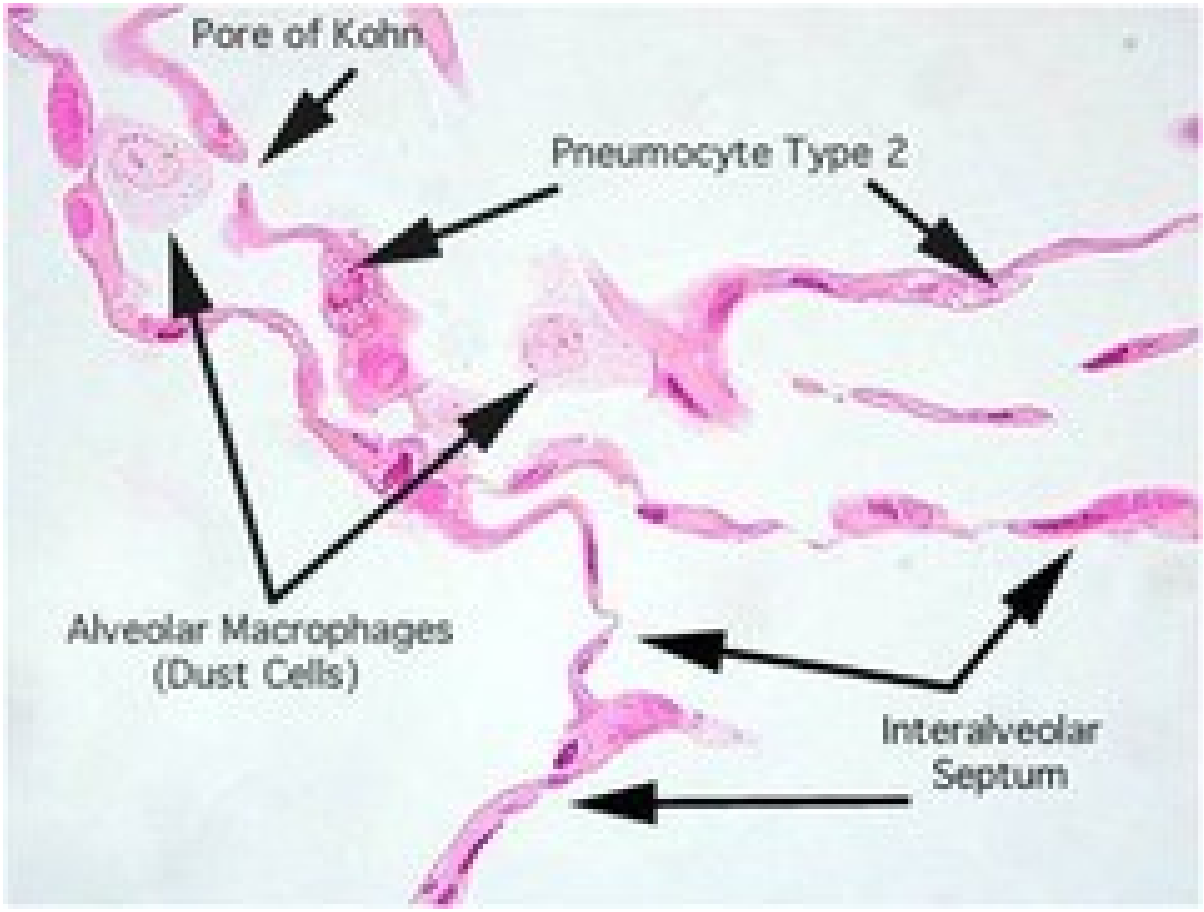

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2

The **interalveolar septum** separating two adjacent alveoli is a very thin structure, consisting of epithelium and connective tissue. Both sides of the adjacent alveolar surfaces are lined by alveolar epithelium composed of type I and II cells. Between these two layers of epithelia is an interstitium containing alveolar **capillaries,** supported in a network of elastic and reticular fibers. Some **fibroblasts** are found in the septum. The alveolar **macrophage** or **dust cells** can be observed in the interstitium of the interalveolar septum or free in the alveolar spaces.

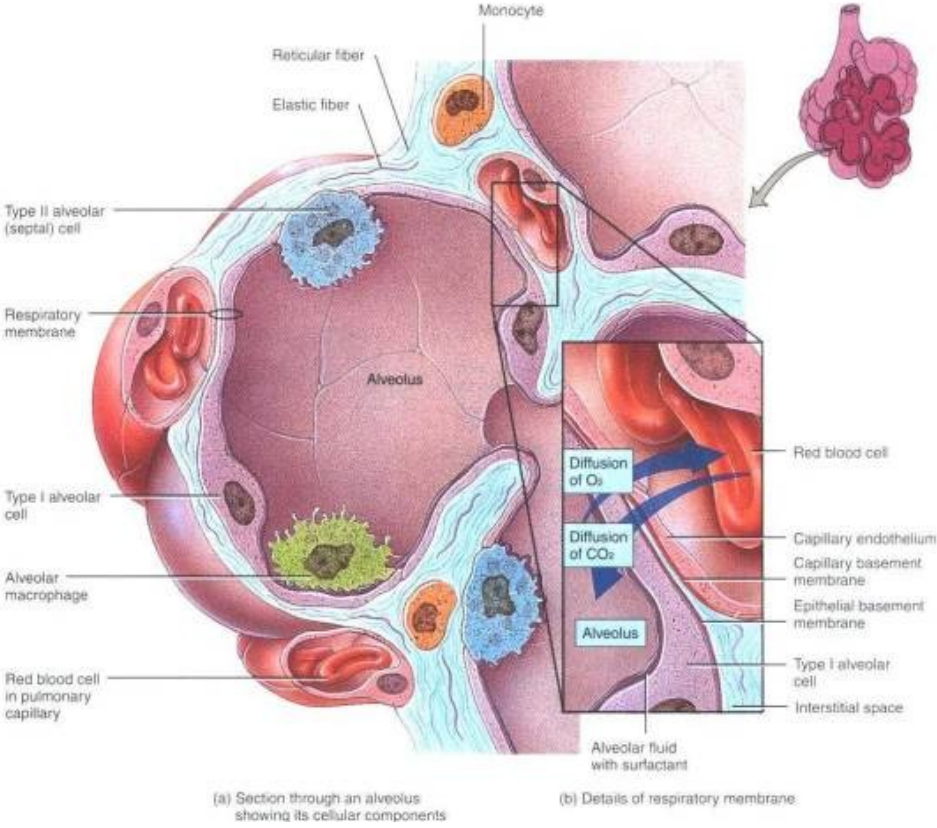


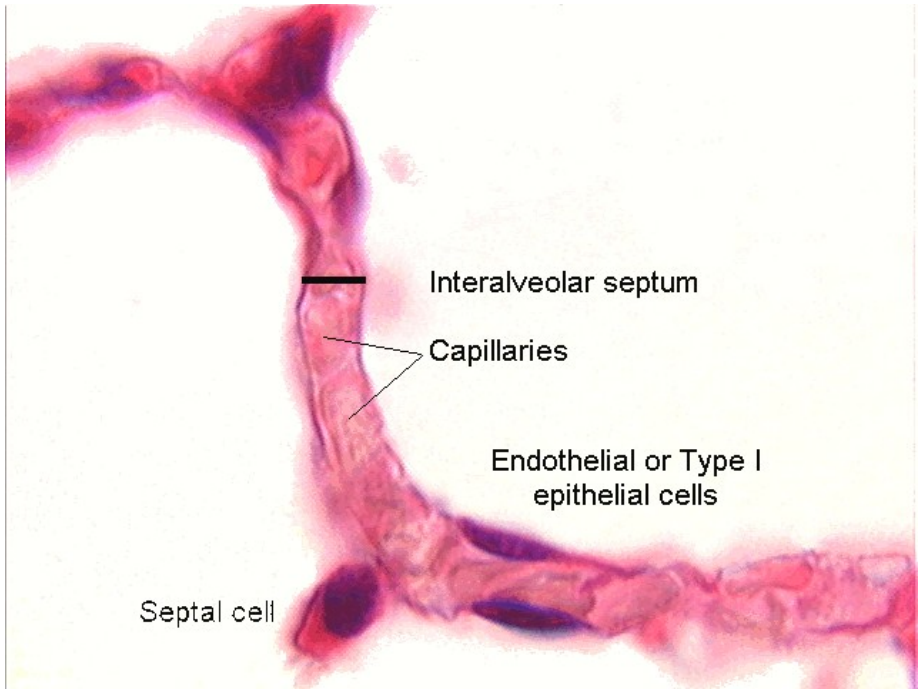


Alveolar pores, 10–15 μm in diameter, may be found in the interalveolar septum, connecting two adjacent alveoli. They may equalize pressure between alveoli, or help to prevent collapse or overdistension of the alveoli if a small bronchiole is occluded.

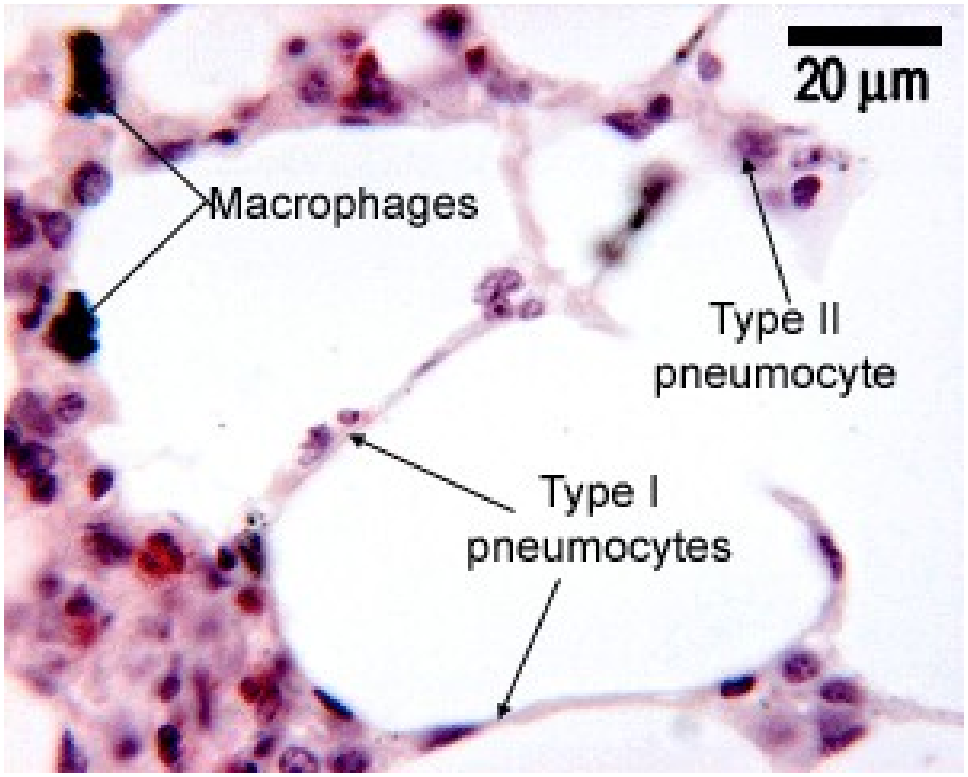


The structures through which gaseous exchange occurs between air in alveoli and blood in alveolar capillaries are known collectively as the *blood–air barrier.* These structures, measuring about 0.1–0.2 μm in thickness, include (1) attenuated cytoplasm of pulmonary epithelial cells; (2) basement membrane of the epithelium; (3) basement membrane of the capillary; and (4) very thin cytoplasm of capillary endothelial cells. To reduce the thickness of the blood–air barrier, the two basement membranes are generally fused into one layer.

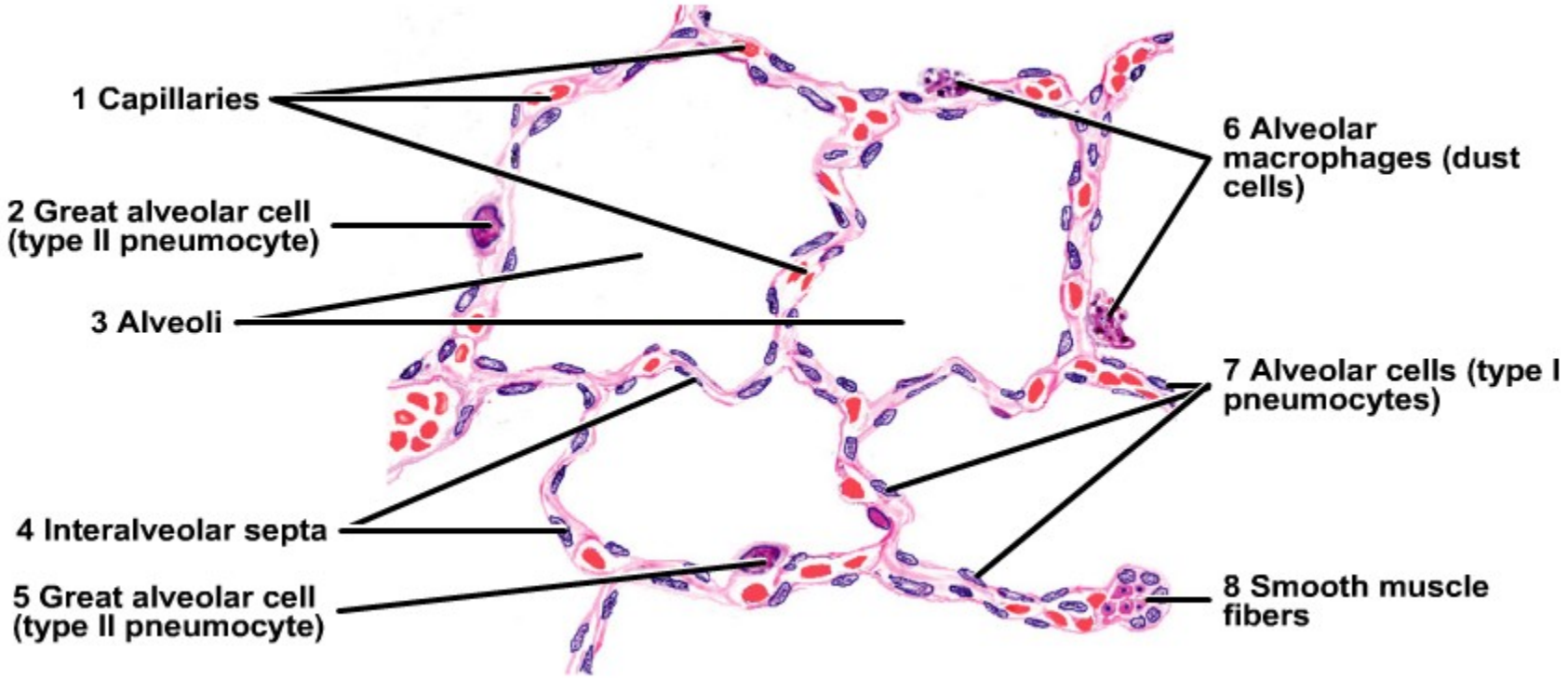




septum. The alveolar **macrophage** or **dust cells** can be observed in the interstitium of the interalveolar septum or free in the alveolar spaces. At the light microscopic level, the dust cells are characterized by enormous numbers of black phagocytosed carbon particles deposited in their cytoplasm. The lungs of heavy smokers appear darker in color because of numerous dust cells with carbon particles.

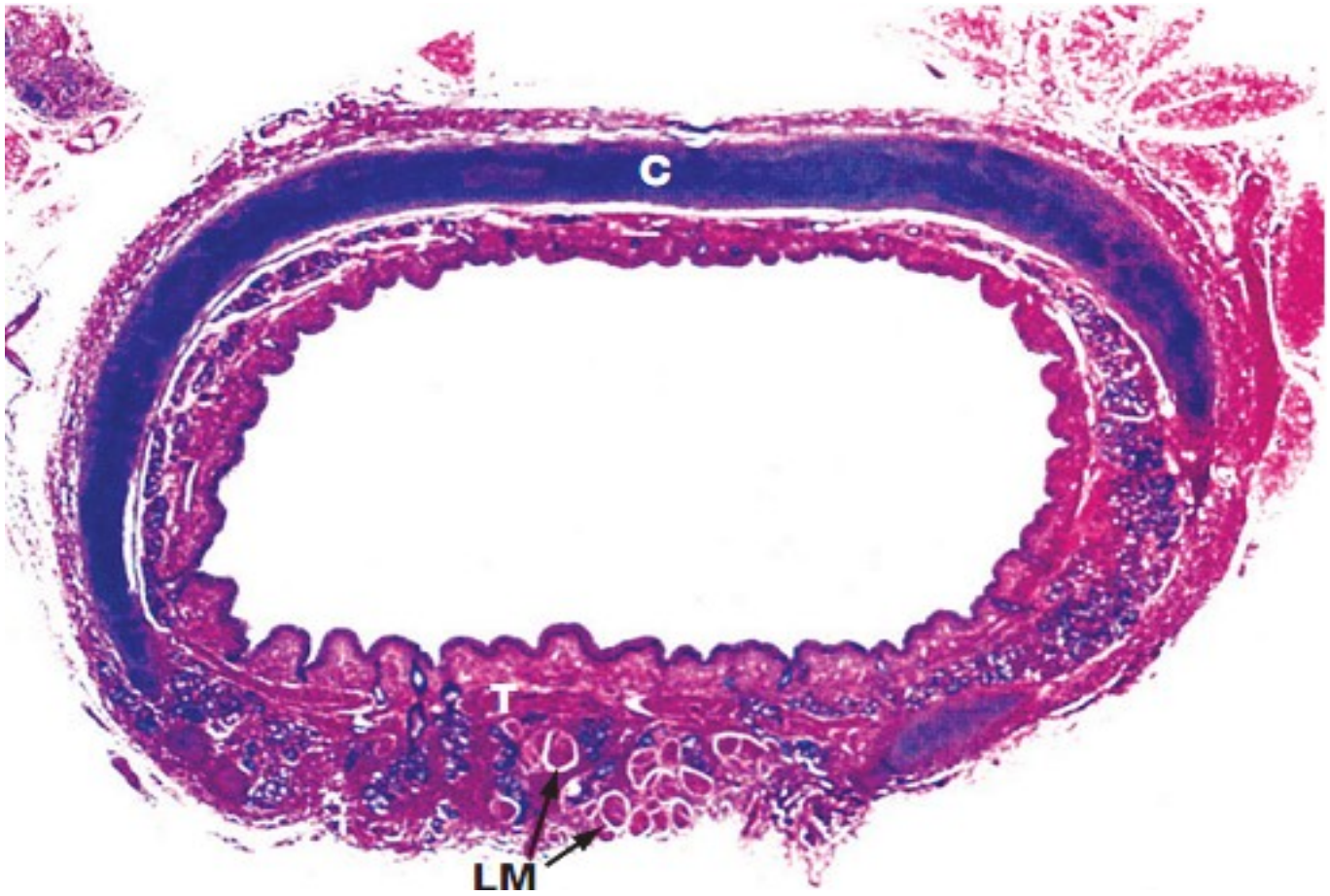


ALVEOLAR WALL AND ALVEOLAR CELLS

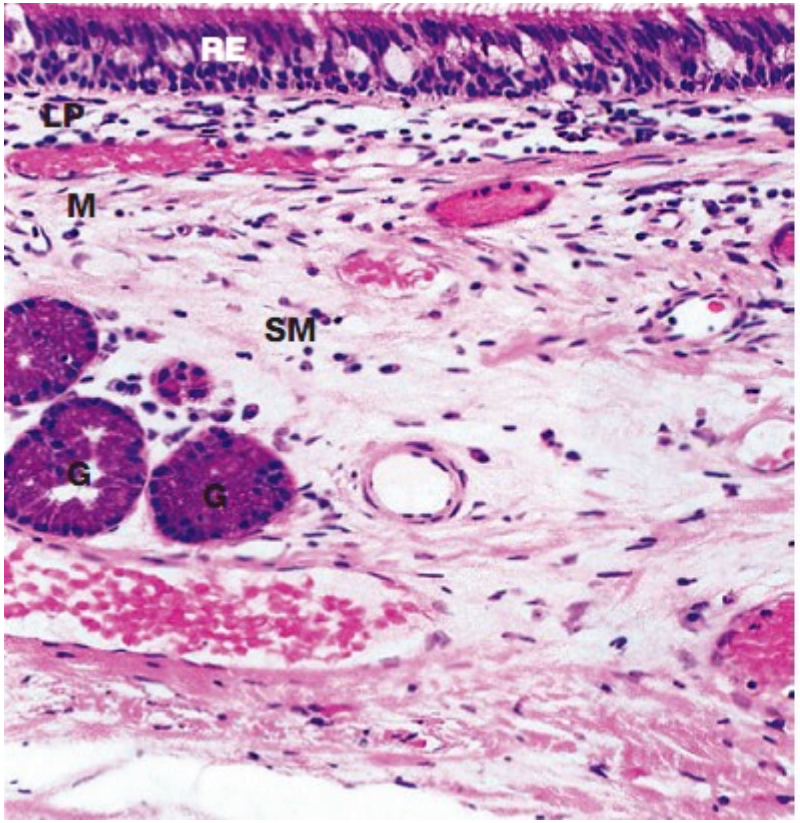


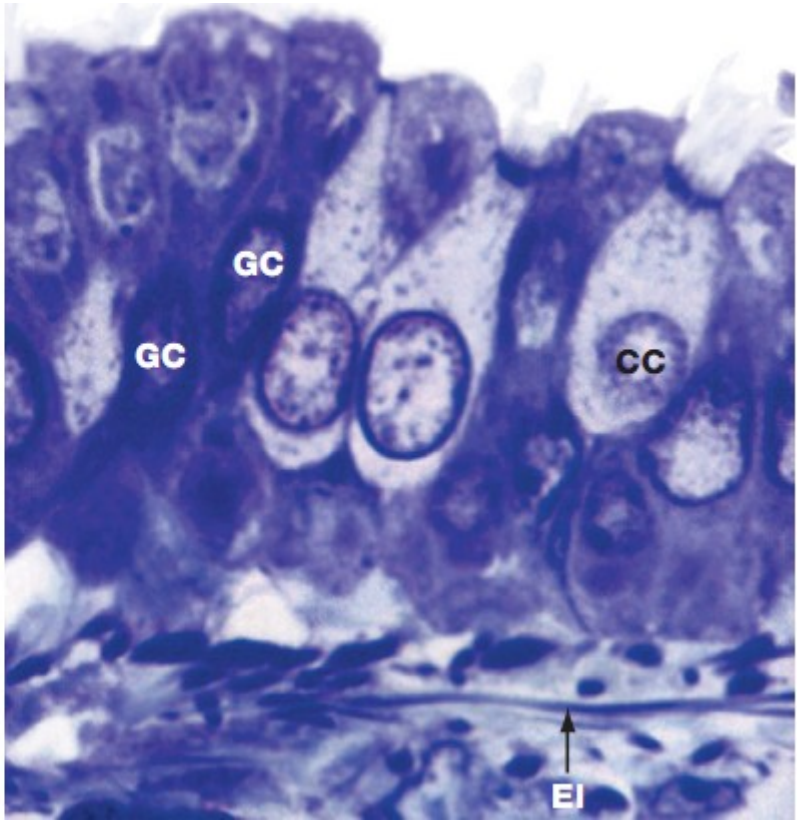

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YUMMY
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Trachea



Primary bronchus





Bronchiole

